Generative Histories, Arithmetic Recombination, and Higher-Dimensional Route Topology

UNNS Research Collective

Manuscript II of the programme *GEO-HD-1* — version 0.4

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Abstract

On the positive integers consider the moves $m \mapsto m(N+1)^2/N$, admissible whenever the label N divides m. A *history* is a finite sequence of such moves. We show that the history, and not the integer it reaches, is the mathematical object that carries the arithmetic and the topology of this system.

Histories ending at one integer are counted exactly by a recursion over the squares dividing the endpoint, with a binomial lower bound along a family. Whether a word of labels is admissible from a given source is decided by a single divisibility $K(w) \mid m$, where $K(w)$ is a product of prime powers read off the valuation walk. From this we obtain a diamond criterion, a saturation theorem, a classification of swap classes for a family of three labels, a block factorisation of the swap graph along the strongly connected components of a feeding digraph, and the exact classification of three-label contents. Complexes of feasible label subsets inside the permutohedron are studied by Alexander duality: a complementary pair of infeasible subsets gives the homology of a sphere, realised by explicit loops and voids.

Histories with different contents reach one endpoint exactly when the content difference lies in the integer kernel of a valuation matrix. We realise every kernel vector, give the exact admissibility threshold of a block rewrite, separate algebraic, exposure and emptiness obstructions, and prove an exposure dichotomy for an infinite family. Finally, an occurrence-event invariant separates histories of one endpoint although the geometry of that endpoint is contractible. Finite-exact computations are kept separate from theorems throughout.

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1 Introduction and contributions

We study the dynamics on the positive integers in which a state m moves to

$$T_N(m) = m\,R(N), \qquad R(N) = \frac{(N+1)^2}{N},$$

whenever the *label* $N \geq 1$ divides m. In the UNNS programme this is the integer grammar SELECT-II [2, 4]; Manuscript I [1] studied the geometry carried by a single state of the programme's arithmetic, and the present manuscript studies what happens *between* states. A finite sequence of such moves is a *history*. The point of view taken here is that the history is a mathematical object in its own right: it has an arithmetic (which words of labels are admissible from which sources), a combinatorics (how many histories end at one integer, how adjacent exchanges connect them) and a topology (cell complexes whose vertices are histories), and none of this is recoverable from the integer at which a history ends.

The four histories of 72. The integer 72 is reached from four complete histories (Figure 1, Figure 2):

$$1 \to 4 \to 16 \to 72, \qquad 1 \to 4 \to 18 \to 72, \qquad 3 \to 16 \to 72, \qquad 10 \to 72.$$

The first two have the same source, content $\{1, 1, 2\}$ and endpoint, and differ by one exchange of adjacent labels; the third and fourth have other contents altogether. The same endpoint hides three contents and three swap classes. The contribution of this manuscript is to show that this phenomenon is governed by exact arithmetic that can be stated and proved, and to chart the topology it produces.

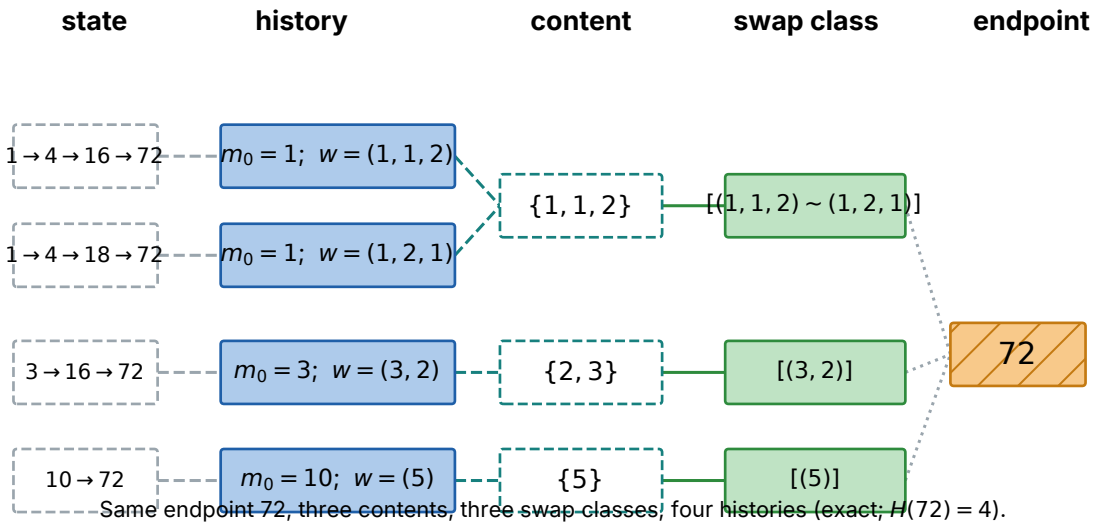


Figure 1: **Objects and equivalence relations (exact, endpoint 72).** The four complete histories ending at 72 (middle left, with their state sequences in dashed boxes), their three contents, the swap classes they fall into, and the common endpoint. Source m_0 and word w are as in Definition 2.4; the swap relation is Definition 5.1.

Contributions. The results are organised in nine groups, HII-1 to HII-9; each is a part of the programme's allocation for Manuscript II and is tagged in the text by the identifiers of the source records it restates (Appendix A).

HII-1 *Complete ancestry and counting* (Section 3). The predecessors of an integer t correspond to the squares $d^2 \mid t$; $H(t) = \sum_{d^2 \mid t} H(t(d-1)/d^2)$ with $H = 1$ on squarefree t, and the recursion terminates because $R(N) \geq 4$ (Theorem 3.1). $H(72) = 4$, $H(72^r) \geq \binom{2r}{r}$ (Proposition 3.4), and the exact values for $r \leq 6$ are finite-exact computations (table 1).

HII-2 *Ordered-word admissibility* (Section 4). A word w is admissible from m iff $K(w) \mid m$, with $K(w) = \prod_p p^{\max(0, \max_j - \sum_{i \leq j} \Delta_p(N_i))}$ and $\Delta_p(N) = 2v_p(N+1) - v_p(N)$ (Theorem 4.2); the diamond criterion, the prefix-set form, the gcd reduction and the saturation criterion $P(C) \mid m$ follow.

HII-3 *Swaps* (Section 5). The swap graph at a source is an induced subgraph of the arrangement graph; classes only merge or appear as the source gains divisors; numerical commutativity of the multipliers never implies admissibility; the contents $(14, 3, 2)$ and $(2, 3, 14)$ lie in different classes of one source and endpoint, and the number of classes of $\{2, 3, 14\}$ is classified for every source (Theorem 5.8).

HII-4 *Feeding, factorisation and components* (Section 6). Non-feeding pairs commute; the class count is a product over the strongly connected blocks of the feeding digraph (Theorem 6.3); an acyclic feeding digraph forces at most one class; three distinct labels split exactly at antipodal deadlocks (Theorem 6.7); $\{2, 3, N\}$ has a split source iff $N \equiv 2, 3 \pmod 6$ (Theorem 6.9).

HII-5 *Feasible-subset geometry and topological obstructions* (Section 7). Alexander duality identifies the homology of the permutohedral route complex with that of a nerve of the infeasible arrangements (Theorem 7.3); lower sets are contractible; a complementary pair of infeasible subsets gives the homology of S^{k-3} (Theorem 7.5; the homotopy type is not proved), with a loop for $\{2, 3, 8, 9\}$ at 72 and a void for $\{2, 3, 8, 9, 14\}$ at 1008.

HII-6 *Higher-dimensional complexes* (Section 8). Commuting swaps give cubes (Lemma 8.1); the maximal cube dimension of 72^r and the swap-class counts are finite-exact; cube dimension is not homological degree; irreducible prime-resource obstructions (Proposition 8.9).

HII-7 *Different-content endpoint relations* (Section 9). Same endpoint iff the content difference lies in the integer kernel of a valuation matrix (Theorem 9.1); every kernel vector is realised (Theorem 9.2); the integer kernel of $\{1, 2, 3\}$ is $\mathbb{Z}(7, -2, -4)$ and gives the histories $4 \to 65536$ of Figure 17.

HII-8 *Block rewrites, exposure and resources* (Section 10). The exact admissibility threshold of a block rewrite (Theorem 10.2); saturation; the trichotomy algebraic/exposure/emptiness; the exposure dichotomy of $2^4 3^j \leftrightarrow 1 \cdot 8^2 3^j$ for all j (Theorem 10.8), and a counterexample showing that a restricted relation set can omit a Graver move (Proposition 10.7).

HII-9 *Occurrence events and state geometry* (Sections 11 and 12). Occurrence-event invariants (births, losses, churn, and the scoped separator $I = [2 \text{ before } 4]$ of Theorem 11.9); the contractibility of the endpoint's move complex shows that no homology of state geometry sees the history components (Proposition 12.1).

What is new in treatment, and what is not claimed. Every theorem of this manuscript restates and re-proves a result of the programme's internal releases (listed in Appendix A); every finite computation not marked CITED was re-run with an independent exact engine written for this manuscript (Appendix B, `scripts/`), and several statements are sharpened or corrected where the re-derivation required it (Section 13). *No claim of novelty or priority is made.* Several of the structures resemble known ones—Mazurkiewicz traces and commutation classes [6, 8, 7], rewriting systems and confluence [9, 10], Markov and Graver bases of toric ideals [11, 12, 13], directed topology of concurrent programs with semaphores [18, 19]—and each such resemblance is pointed out where it occurs and is *only* a resemblance: we prove no equivalence with any of them, and the swap relation here depends on the state through divisibility and is not a trace congruence (Remark 5.4). The programme's same-author caveat applies: all proofs and code originate from one author-collective; no independent human review is claimed. The claim C3 of the programme is not made.

Model boundaries. The manuscript uses only two models of the programme: the integer grammar SELECT-II (model M0) and the history, route and rewrite complexes built on it (model M6). The rational dynamics FIELD-I and its occurrence geometry [3] are *not* used, and SELECT-II is never extended to rational states; the correspondence between the integer move complex of an endpoint and the history space is discussed only as a non-relation in Section 12. The two dynamics are different maps and nothing here identifies them (the programme's erratum on this point is recorded in Appendix A).

Status vocabulary. Statements are labelled PROVED (a complete argument is given in this manuscript), PROVED-SCOPED (proved under stated hypotheses, with the scope part of the statement), FINITE-EXACT (an exhaustive exact computation in a stated range, no claim beyond it), COUNTEREXAMPLE, OPEN and CITED (taken from a released source without re-proof). Finite-exact tables are never extrapolated.

Organisation. Section 2 fixes the grammar and the vocabulary. Sections 3 to 6 treat counting, admissibility, swaps and their component structure. Sections 7 and 8 treat the topology of the route complexes; Sections 9 and 10 the relations between histories with different contents; Sections 11 and 12 occurrence events and state geometry; Section 13 the conclusions and open problems. The appendices give the source crosswalk, the complete examples, the construction of the complexes, the kernel and rewrite calculations, and the supplementary proofs.

2 Grammar, notation and history spaces

2.1 The grammar

Definition 2.1 (SELECT-II)**.** For an integer *label* $N \geq 1$ put $R(N) = (N+1)^2/N$ and $T_N(m) = m\,R(N)$. The label N is *admissible at* the integer $m \geq 1$ if $T_N(m) \in \mathbb{Z}$.

Lemma 2.2 (admissibility is divisibility)**.** *For a positive integer m, $T_N(m) \in \mathbb{Z}$ if and only if $N \mid m$.*

Proof. $\gcd(N, (N+1)^2) = 1$, so $N \mid m(N+1)^2$ iff $N \mid m$. □

Lemma 2.3 (multipliers)**.** *$R(N) = N + 2 + 1/N$; $R(1) = 4$, R is strictly increasing on integers $N \geq 1$, and hence $R(N) \geq 4$ with equality only for $N = 1$. In particular $N \mapsto R(N)$ is injective.*

Proof. $R(N+1) - R(N) = 1 - \frac{1}{N(N+1)} > 0$. □

For a prime p let v_p be the p-adic valuation and put

$$\Delta_p(N) = v_p\big(R(N)\big) = 2v_p(N+1) - v_p(N),$$

so at most one of $v_p(N), v_p(N+1)$ is nonzero and $\Delta_p(N) \geq -v_p(N)$ with equality unless $p \mid N+1$. A prime is *active* for a set of labels if it divides one of them. All valuations of $R(N)$ vanish outside the primes dividing $N(N+1)$.

Definition 2.4 (history)**.** A *history* is a pair $h = (m_0; w)$ with $m_0 \in \mathbb{Z}_{\geq 1}$ the *source* and $w = (N_1, \ldots, N_l)$ a word of labels such that $m_j = T_{N_j}(m_{j-1})$ is an integer for $j = 1, \ldots, l$, i.e. $N_j \mid m_{j-1}$. The integers $m_0, \ldots, m_l$ are the *states* of h, m_l is the *endpoint*, l the length, and the multiset of letters of w is the *content*. The history is *complete* if m_0 is squarefree (equivalently, by Theorem 3.1(b), m_0 has no predecessor, so the history cannot be extended backwards).

By Lemma 2.3 states increase strictly, so a history determines the set of its states, and $m_l = m_0 \prod_j R(N_j)$ depends only on the source and the content.

2.2 Vocabulary: state, history, content, endpoint, swap class, rewrite class

The words are used in a fixed sense throughout.

- A *state* is an integer m. A *history* is a path $(m_0; w)$. Its *content* is the multiset of its labels and its *endpoint* the last state. Two histories have the same endpoint from the same source iff they have the same multiplier $\prod R(N_j)$ (Section 9).
- Two histories of the same source are *swap-adjacent* if their words differ by exchanging two adjacent *distinct* labels (Definition 5.1); the connected components of the resulting graph are *swap classes*. Swap classes lie inside content classes, which lie inside endpoint classes.
- A *block rewrite* replaces a contiguous block of a word by another word of equal multiplier, possibly with a different content (Definition 10.1); the components of swaps and block rewrites together are *rewrite classes*.

Figure 1 displays the nesting for the endpoint 72. A function of the endpoint alone sees none of the finer relations; the manuscript's results are statements about the finer relations.

2.3 Valuation walks

For an integer m write $u(m) = (v_p(m))_p$. A step with label N requires $u_p \geq v_p(N)$ for all p and changes u by $\Delta(N) = (\Delta_p(N))_p$. A history is therefore a lattice walk in the nonnegative orthant with step set $\{\Delta(N)\}$: p is a resource of capacity u_p, a label divisible by p consumes it, and a label N with $p \mid N+1$ produces it. The total displacement of a walk is determined by the content, so it is the *route* of the walk—not its endpoint—that distinguishes histories. This is the picture behind Theorem 4.2 below; it is inherited arithmetic (record `D-W`).

Source record: `D-W` (`CW-0337`); status PROVED; treatment FULL.

Remark 2.5 (model boundaries). Only integer states and the grammar of Definition 2.1 are used. Rational states, the collective map of the programme's FIELD-I, the occurrence geometry of rational states and the move complexes of rational states are not used; where an integer move complex enters (Section 12) it is the complex of an integer endpoint, which is not a rational extension.

3 Complete ancestry and exact counting

3.1 Predecessors

Theorem 3.1 (predecessors and the count recursion)**.** *Let $t \geq 1$ be an integer.*

(a) $m \xrightarrow{N} t$ (that is, $T_N(m) = t$ with $N \mid m$) if and only if $d := N+1$ satisfies $d^2 \mid t$ and $m = t(d-1)/d^2$. The correspondence $d \leftrightarrow (N, m)$ is a bijection.

(b) Hence t has exactly $\#\{d \geq 2 : \ d^2 \mid t\}$ incoming edges, one for each square divisor $d^2 \geq 4$, and t has none iff t is squarefree.

(c) Let $H(t)$ be the number of complete histories ending at t. Then $H(t) = 1$ for squarefree t and, for every other t,

$$H(t) = \sum_{d \geq 2, \ d^2 \mid t} H\left(\frac{t(d-1)}{d^2}\right).$$

(d) Every history ending at t from a source m_0 has length at most $\log_4(t/m_0)$; in particular $H(t)$ is finite.

Proof. (a) If $m = Nk$ and $T_N(m) = t$ then $t = k(N+1)^2$, so $(N+1)^2 \mid t$ and $m = Nt/(N+1)^2$. Conversely, if $d^2 \mid t$ and $N = d-1$, put $m = Nt/d^2 = N \cdot (t/d^2)$; then $N \mid m$ and $T_N(m) = m\,d^2/N = t$. The two maps are mutually inverse, and N is determined by $d = N+1$. (b) is the count of the d of (a); t has a square divisor $d^2 \geq 4$ iff t is not squarefree. (c) A complete history of a non-squarefree t has positive length, because its source is squarefree. Removing its last edge, which is an incoming edge of t of the form in (a), leaves a complete history ending at the predecessor m. The map (history) $\mapsto$ (last edge, remaining history) is a bijection onto pairs consisting of an incoming edge and a complete history of its tail. For squarefree t only the empty history is complete, since t has no incoming edge. (d) By Lemma 2.3, $t = m_0 \prod R(N_j) \geq 4^l m_0$. □

Source record: `D-T0` (`CW-0336`); status PROVED; treatment FULL.

Remark 3.2. The recursion is the definition of H as a path count in the *predecessor graph* of t: the directed graph on all ancestors of t whose edges are the incoming edges of (a). It is acyclic because states strictly increase along edges, so the recursion terminates, and Figure 3 shows it for $t = 72$.

Example 3.3 (the four histories of 72)**.** Since $72 = 2^3 3^2$, the squares dividing 72 are $d^2 = 4, 9, 36$ ($d = 2, 3, 6$; $16 \nmid 72$ and $144 > 72$), with predecessors $72 \cdot 1/4 = 18$ (label $N = 1$: $18 \cdot 4 = 72$), $72 \cdot 2/9 = 16$ (label 2: $16 \cdot \frac{9}{2} = 72$) and $72 \cdot 5/36 = 10$ (label 5: $10 \cdot \frac{36}{5} = 72$). Next, $18 = 2 \cdot 3^2$ has the single square 9 with predecessor $18 \cdot 2/9 = 4$ (label 2: $4 \cdot \frac{9}{2} = 18$); 16 has squares 4 and 16 ($d = 2, 4$), with predecessors $16/4 = 4$ (label 1) and $16 \cdot 3/16 = 3$ (label 3: $3 \cdot \frac{16}{3} = 16$); 4 has the square 4 with predecessor 1 (label 1); and $10, 3, 1$ are squarefree. Hence

$$H(18) = H(4) = H(1) = 1, \quad H(16) = H(4) + H(3) = 2, \quad H(10) = 1, \quad H(72) = 1+2+1 = 4.$$

Reading the four paths from the sources $1, 1, 3, 10$ gives

$$1 \xrightarrow{1} 4 \xrightarrow{1} 16 \xrightarrow{2} 72, \qquad 1 \xrightarrow{1} 4 \xrightarrow{2} 18 \xrightarrow{1} 72, \qquad 3 \xrightarrow{3} 16 \xrightarrow{2} 72, \qquad 10 \xrightarrow{5} 72,$$

with words $(1,1,2), (1,2,1), (3,2), (5)$ and contents $\{1,1,2\}, \{1,1,2\}, \{2,3\}, \{5\}$. The first two differ by exchanging the adjacent distinct labels $1, 2$ at the states $4 \to 16$, $4 \to 18$; the states 16 and 18 are the two intermediate states of the diamond $4 \to 16, 18 \to 72$ (Figure 2). (All sources are squarefree, so all four histories are complete.)

3.2 Growth

Proposition 3.4 (lower bound for 72^r)**.** *For every $r \geq 1$, $H(72^r) \geq \binom{2r}{r}$.*

Proof. Start from source 1. Apply the label 1 exactly r times, reaching $4^r = 2^{2r}$ (every state is divisible by the label 1). We show that every arrangement of r labels 1 and r labels 2 is admissible from 4^r. At a point where a labels 1 and b labels 2 have been applied so far, the 2-adic valuation of the state is $2r + 2a - b$, since a label 1 adds 2 and a label 2 changes v_2 by -1 (the valuation of $R(2) = 9/2$ is -1 at 2). Before the $(b+1)$-st label 2 we have $b \leq r-1$, so $v_2 \geq 2r - b \geq r + 1 \geq 1$ and 2 divides the state. The endpoint of any such arrangement

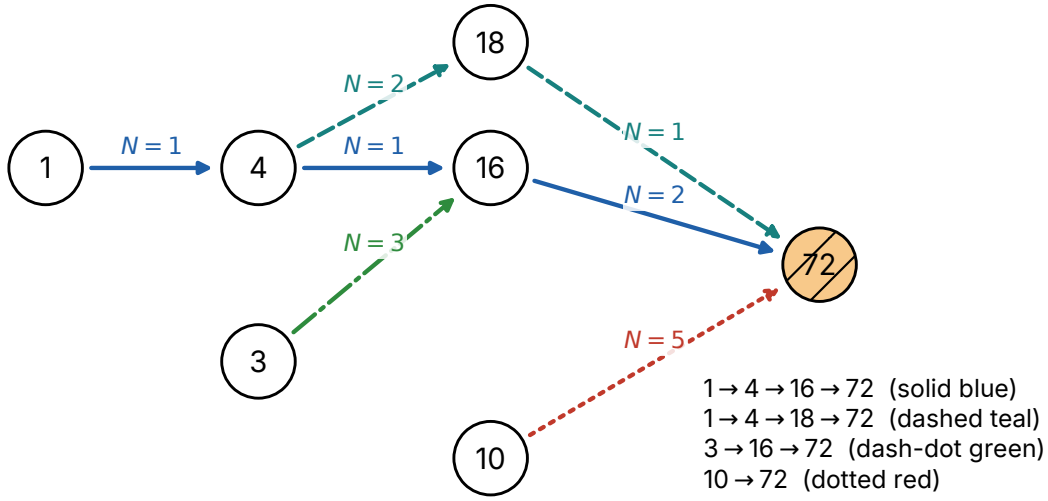


Figure 2: **The four complete histories ending at 72 (exact).** States are nodes, labels N are edge labels. Source 1: $1 \to 4 \to 16 \to 72$ (solid blue) and $1 \to 4 \to 18 \to 72$ (dashed teal); source 3: $3 \to 16 \to 72$ (dash-dot green); source 10: $10 \to 72$ (dotted red). Theorem illustrated: Theorem 3.1; Example 3.3.

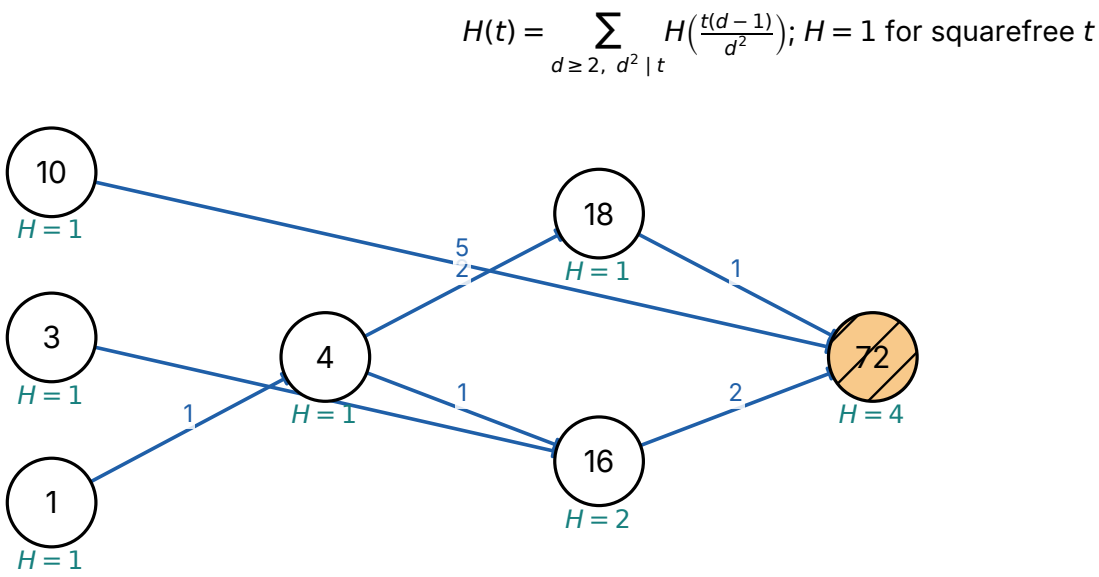


Figure 3: **Predecessor graph of 72 and the count recursion (exact).** Nodes are the ancestors of 72; an edge $s \to t$ carries its label N; the number under a node is H of that node, computed by the recursion of Theorem 3.1(c). $H(72) = 4$.

r	1	2	3	4	5	6
$H(72^r)$	4	100	3 601	194 864	14 177 856	1 258 011 667
$\binom{2r}{r}$	2	6	20	70	252	924

Table 1: **Counts of complete histories ending at 72^r (finite-exact).** Values of H from the recursion Theorem 3.1(c), computed exactly in this work and agreeing with the programme's records; the bound is Proposition 3.4. These are finite computations for $r \le 6$: no growth law in r is claimed (it is open, Problem 3.6).

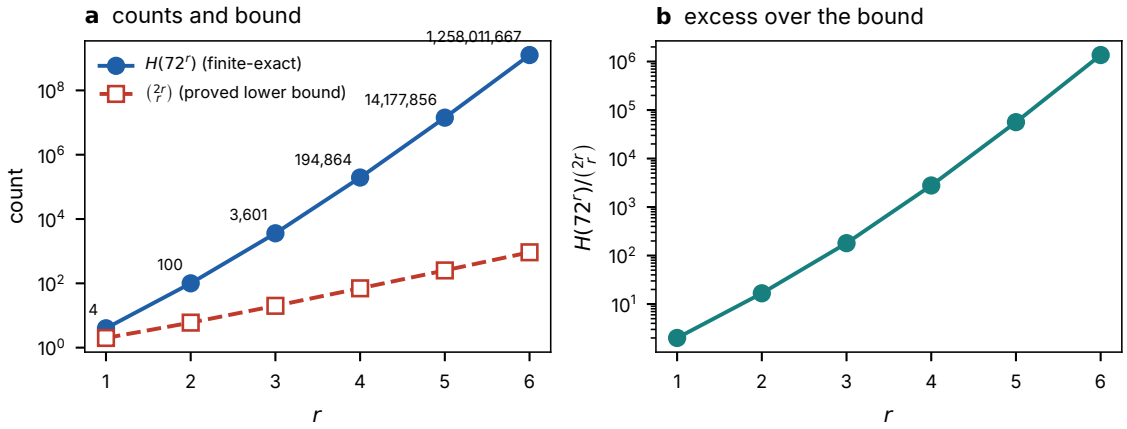


Figure 4: **Growth of $H(72^r)$ and its lower bound (finite-exact values, proved bound).** (a) Counts for $r = 1, \dots, 6$ (circles, table 1) against $\binom{2r}{r}$ (squares, Proposition 3.4). (b) The ratio; the bound is far from sharp.

is $4^r \cdot 4^r \cdot (9/2)^r = 72^r$ (the multiplier is $R(1)^r R(1)^r R(2)^r = 4^{2r}(9/2)^r = (16 \cdot 9/2)^r = 72^r$). Distinct arrangements are distinct words from one source, hence distinct histories; there are $\binom{2r}{r}$ of them, and the whole history $(1^r) \cdot$ (arrangement) is from source 1, which is squarefree, so these are complete. □

Source record: `D-T2` (`CW-0339`); status PROVED; treatment FULL.
The figure above is Figure 4; its caption states the exactness of what is drawn.

Remark 3.5. The values of table 1 for $r \le 4$ were additionally confirmed by enumerating the histories explicitly; the values for $r = 5, 6$ rest on the recursion, which counts paths in an acyclic graph and whose correctness is Theorem 3.1.

Problem 3.6 (open: growth)**.** Determine the asymptotics of $H(72^r)$ (and of the maximum cube dimension, Section 8). Only the lower bound of Proposition 3.4 and the finite values of table 1 are known here.

Source record: `D-Q2` (`CW-0514`); status OPEN; treatment OPEN.

4 Ordered-word admissibility and valuation thresholds

Fix a word $w = (N_1, \dots, N_l)$ of labels and put $S_j(p) = \sum_{i \le j} \Delta_p(N_i)$, so that $m \prod_{i \le j} R(N_i)$ has p-adic valuation $v_p(m) + S_j(p)$.

Definition 4.1 (threshold). $b_p(w) = \max\big(0, \max_{1\le j\le l}(-S_j(p))\big)$ and $K(w) = \prod_p p^{\,b_p(w)}$.

Only the finitely many primes dividing some $N_i(N_i+1)$ contribute.

Theorem 4.2 (valuation-threshold theorem). *w is admissible from m if and only if $K(w) \mid m$. Hence $K(w)$ is the smallest source from which w is admissible, and the admissible sources are exactly the multiples of $K(w)$.*

Proof. Put $x_j = m\prod_{i\le j} R(N_i) \in \mathbb{Q}_{>0}$ (so $x_0 = m$). We claim w is admissible from m iff $x_j \in \mathbb{Z}$ for $j = 0, \dots, l$. If w is admissible the x_j are the states. Conversely suppose all x_j are integers; then $x_j = x_{j-1}(N_j+1)^2/N_j$ with $x_{j-1} \in \mathbb{Z}$, so $x_j \in \mathbb{Z}$ gives $N_j \mid x_{j-1}$ by Lemma 2.2, i.e. the step is admissible. Next, a positive rational is an integer iff all its p-adic valuations are nonnegative, and $v_p(x_j) = v_p(m) + S_j(p)$. Therefore w is admissible iff $v_p(m) + S_j(p) \ge 0$ for all p and j, i.e. iff $v_p(m) \ge \max_j(-S_j(p))$ for all p; since $v_p(m) \ge 0$ this is $v_p(m) \ge b_p(w)$ for all p, i.e. $K(w) \mid m$. □

Source record: `D-T7` (`CW-0344`); status PROVED; treatment FULL.

Source record: `D6-T1` (`CW-0365`); status PROVED; treatment FULL.

For a finite multiset S of labels put $R(S) = \prod_{x\in S} R(x)$ and

$$\kappa(S) = \prod_p p^{\max(0,-\sum_{x\in S}\Delta_p(x))} = \operatorname{den} R(S),$$

the denominator of $R(S)$ in lowest terms (the primes with $v_p(R(S)) < 0$, to the power $-v_p$). A multiset S is *feasible at m* if $\kappa(S) \mid m$, i.e. if $mR(S) \in \mathbb{Z}$.

Corollary 4.3 (prefix-set form). *$K(w) = \operatorname{lcm}_{1\le j\le l}\ \kappa\big(\{N_1, \dots, N_j\}\big)$. Consequently w is admissible from m iff every prefix multiset of w is feasible at m; and $K(w)$ depends on w only through its prefix multisets.*

Proof. $S_j(p) = \sum_{x\in\{N_1..N_j\}}\Delta_p(x)$ depends only on the prefix multiset, so $b_p(w) = \max_j \max(0, -S_j(p))$ is the largest exponent of p among the $\kappa(\text{prefix}_j)$, and the maximum of exponents is the exponent of the lcm. □

Source record: `(P) Prefix-set form` (`CW-0366`); status PROVED; treatment FULL.

For a single label $\kappa(\{N\}) = N$, because $\Delta_p(N) = -v_p(N)$ when $p \mid N$ and $\Delta_p(N) \ge 0$ otherwise. For a multiset C let $P(C) = \prod_{x\in C} x$ (with multiplicity).

Corollary 4.4 (gcd reduction). *For every arrangement w of C, $K(w) \mid P(C)$. Hence w is admissible from m iff it is admissible from $\gcd(m, P(C))$.*

Proof. Since $\Delta_p(N) \ge -v_p(N)$ we have $-S_j(p) \le \sum_i v_p(N_i) = v_p(P(C))$, so $b_p(w) \le v_p(P(C))$. If $K(w) \mid P(C)$ then $K(w) \mid m \iff K(w) \mid \gcd(m, P(C))$. □

Source record: `(G) Dependence on the source` (`CW-0367`); status PROVED; treatment FULL.

Proposition 4.5 (diamond criterion). *For labels $A \ne B$: (a) $K\big((A,B)\big) = AB/\gcd(B, (A+1)^2)$, so (A,B) is admissible from m iff $\frac{AB}{\gcd(B,(A+1)^2)} \mid m$; (b) both (A,B) and (B,A) are admissible from m iff $AB \mid m$; (c) if a history passes through the state x immediately before two adjacent distinct labels A, B, then exchanging them gives an admissible history iff $AB \mid x$.*

Proof. (a) By Lemma 2.2, $A \mid m$, say $m = Ak$, and then $B \mid T_A(m) = k(A+1)^2$. Put $g = \gcd(B, (A+1)^2)$, $B = gB'$, $(A+1)^2 = gE$ with $\gcd(B', E) = 1$. Then $B \mid kgE$ iff $B' \mid kE$ iff $B' \mid k$. Hence $m = Ak$ with $B' \mid k$, i.e. $AB' \mid m$, and $AB' = AB/g$. (b) With $g_1 = \gcd(B, (A+1)^2)$ and $g_2 = \gcd(A, (B+1)^2)$, both orders are admissible iff $\operatorname{lcm}(AB/g_1, AB/g_2) = AB/\gcd(g_1, g_2)$ divides m (because g_1, g_2 divide AB, $\operatorname{lcm}(X/g_1, X/g_2) = X/\gcd(g_1, g_2)$). Here $g_1 \mid (A+1)^2$ and $g_2 \mid A$, so $\gcd(g_1, g_2) \mid \gcd((A+1)^2, A) = 1$. (c) The history already admits (A, B) from x, so the exchange is admissible iff (B, A) is admissible from x, iff (by (b)) x is a multiple of both $K((A, B))$ and $K((B, A))$, i.e. of AB; and the remainder of the word starts from the same state $xR(A)R(B)$ in both orders. □

Source record: `D-T1` (`CW-0338`); status PROVED; treatment FULL.

Proposition 4.6 (all arrangements; saturation)**.** *For a finite multiset C of labels and a source m: every arrangement of C is admissible from m if and only if $P(C) \mid m$. In particular, if $P(C) \mid m$ then every arrangement is admissible and the swap graph of C at m is connected (Definition 5.1).*

Proof. ($\Leftarrow$) By Corollary 4.4, $K(w) \mid P(C) \mid m$ for every arrangement w. ($\Rightarrow$) Fix a prime p and an arrangement that lists first all labels divisible by p. These have $\Delta_p(N) = -v_p(N)$ (since $p \nmid N+1$), so after the last of them $S_j(p) = -\sum_i v_p(N_i) = -v_p(P(C))$, and admissibility forces $v_p(m) \geq v_p(P(C))$. For connectivity: the arrangement graph of a multiset under exchanges of adjacent distinct letters is connected, and with all arrangements admissible it is the swap graph. □

Remark 4.7 (what is *not* claimed). Proposition 4.6 does *not* say that a connected swap graph forces $P(C) \mid m$: the content $\{2, 3, 14\}$ has $P = 84$ and a connected swap graph at the source 28 (Theorem 5.8). Saturation is a sufficient resource condition for connectivity, not a necessary one; the true thresholds are the path thresholds of Theorem 5.5.

Source record: `D6-T2` (`CW-0368`); status PROVED; treatment FULL.

Example 4.8 (the thresholds of $\{2, 3, 14\}$)**.** Let $C = \{2, 3, 14\}$. For $w = (14, 3, 2)$ the prefix multisets are $\{14\}, \{14, 3\}, \{14, 3, 2\}$ with $R = \frac{225}{14}$, $\frac{225}{14} \cdot \frac{16}{3} = \frac{600}{7}$, $\frac{600}{7} \cdot \frac{9}{2} = \frac{2700}{7}$, whose denominators are $14, 7, 7$; so $K(14, 3, 2) = \operatorname{lcm}(14, 7, 7) = 14$. For $w = (2, 3, 14)$ the prefix products are $\frac{9}{2}$, 24, $\frac{2700}{7}$ with denominators $2, 1, 7$ and $K = 14$. For $(14, 2, 3)$ the second product is $\frac{225}{14} \cdot \frac{9}{2} = \frac{2025}{28}$, so $K = \operatorname{lcm}(14, 28, 7) = 28$; likewise $K(2, 14, 3) = 28$, and $K(3, 14, 2) = K(3, 2, 14) = 21$ (denominators $3, 7, 7$ and $3, 1, 7$). The six thresholds are $14, 28, 21, 21, 28, 14$ (Figures 5 and 6). From $m = 14$ the two arrangements with threshold 14 are admissible and both end at $14 \cdot \frac{2700}{7} = 5400$.

5 Swaps

5.1 The swap graph

Definition 5.1 (arrangement graph, swap graph, swap class)**.** Let C be a finite multiset of labels. The *arrangement graph* $\mathcal{G}(C)$ has as vertices the distinct arrangements (orderings) of C, two being joined if they differ by exchanging two adjacent *distinct* letters. For a source m let $\mathcal{A}(m) = \{w : K(w) \mid m\}$ be the set of arrangements admissible from m. The *swap graph*

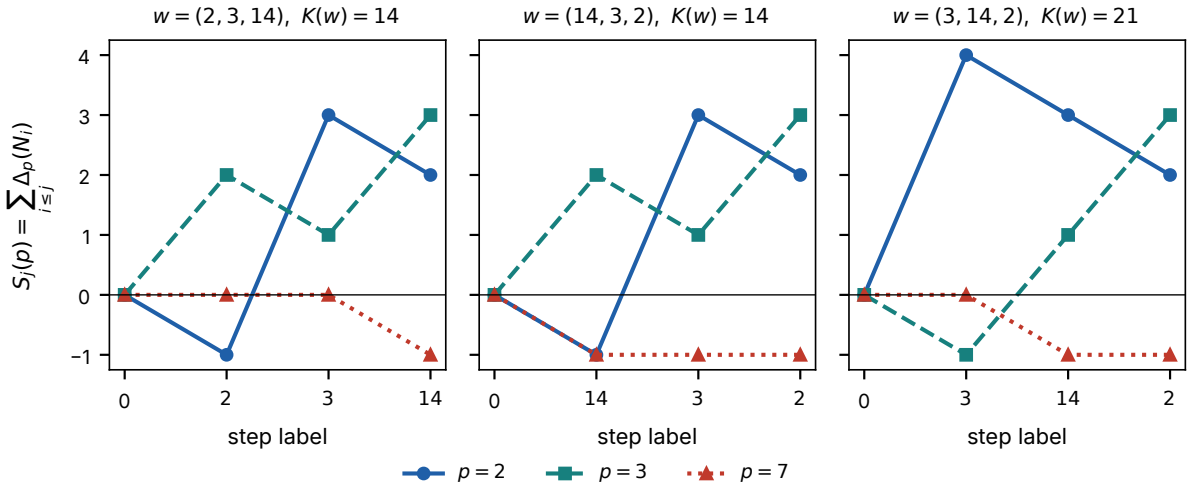


Figure 5: **Valuation trajectories of three words of the content $\{2, 3, 14\}$ (exact).** The curves are $S_j(p)$ for $p = 2$ (solid, circles), 3 (dashed, squares) and 7 (dotted, triangles); $K(w) = \prod_p p^{\max(0, -\min_j S_j(p))}$ is printed above each panel: $14 = 2 \cdot 7$ for $(2, 3, 14)$ and $(14, 3, 2)$ (minima -1 at $p = 2$ and -1 at $p = 7$) and $21 = 3 \cdot 7$ for $(3, 14, 2)$. Theorem illustrated: Theorem 4.2.

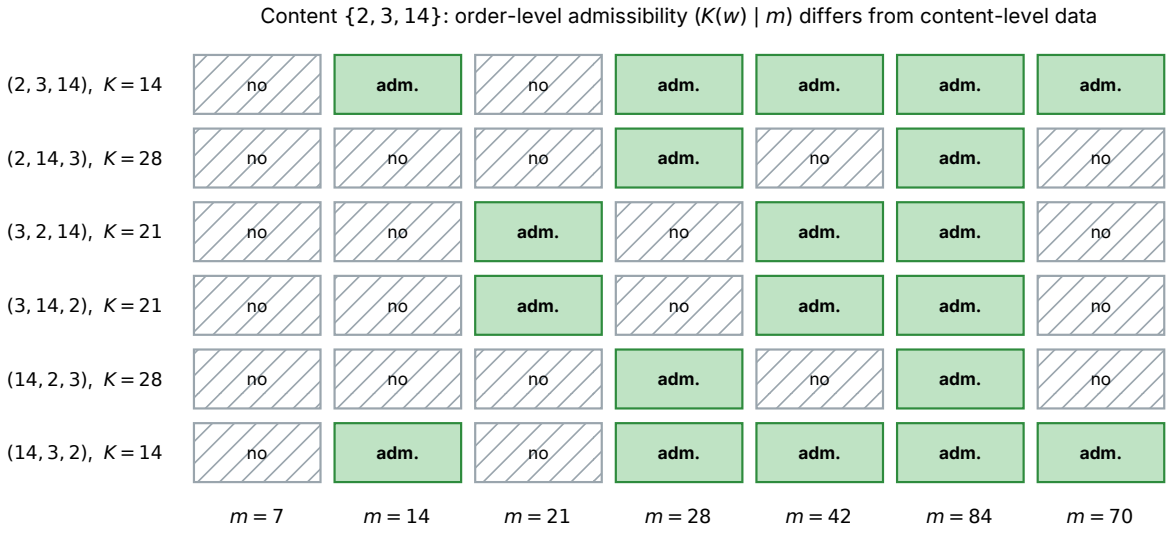


Figure 6: **Content-level versus order-level admissibility for $\{2, 3, 14\}$ (exact).** Rows: the six arrangements with their thresholds K; columns: sources. A cell is filled and marked "adm." iff $K \mid m$. The same content is wholly inadmissible at $m = 7$, partly admissible at $m = 14$ and $m = 21$, and fully admissible among the sources shown only at $m = 84 = P(C)$ (Proposition 4.6); it is connected for $m = 28$ (Theorem 5.8) although $84 \nmid 28$.

$\mathrm{Sw}(C,m)$ is the subgraph of $\mathcal{G}(C)$ induced on $\mathcal{A}(m)$; its edges are the *legal swaps*, and its components are the *swap classes* of C at m. Write $\nu(C,m)$ for their number.

A vertex of $\mathrm{Sw}(C,m)$ is the history $(m;w)$, and a legal swap is an exchange of adjacent distinct labels of a history that produces an admissible history; the endpoint is unchanged because $m\prod R(N_i)$ does not depend on the order.

Proposition 5.2 (the swap graph is an induced subgraph)**.** *The graph on the histories with source m and content C, joined by legal swaps, equals* $\mathrm{Sw}(C,m)$*. In particular two admissible arrangements adjacent in $\mathcal{G}(C)$ are always joined by a legal swap.*

Proof. An edge between two admissible words that differ by an adjacent exchange is, by definition, a legal swap; conversely a legal swap joins two admissible words differing by an adjacent exchange. □

Source record: `D-T7'` (`CW-0345`); status PROVED; treatment FULL.

Proposition 5.3 (numerical commutativity is not admissibility)**.** *For all labels A,B the multipliers commute, $R(A)R(B)=R(B)R(A)$, so both orders of A,B have the same endpoint from every source at which both are admissible. Nevertheless the exchange is legal at the state x iff $AB\mid x$ (Proposition 4.5), which is a condition on x, not on the multipliers. For instance $(2,3)$ is admissible from 2 ($2\to 9\to 48$) while $(3,2)$ is not ($3\nmid 2$).*

Remark 5.4 (resemblance to traces and to concurrency). Adjacent exchanges of commuting letters recall the commutation classes of a trace monoid [8, 6, 7], and the picture of p as a resource of capacity u_p recalls processes with counting semaphores and the directed topology of their state spaces [18, 19]. The resemblance is only that. In a trace monoid independence is a fixed relation on letters; here A,B are interchangeable at x iff $AB\mid x$, which depends on the state, and the classes are not the classes of a congruence on a free monoid. No equivalence with either theory is proved or claimed.

5.2 Thresholds of connection; monotonicity

Theorem 5.5 (connection thresholds and the divisibility filtration)**.** *Let C be a multiset and w,w' arrangements of C.*

(a) w and w' lie in one swap class at the source m iff there is a path γ from w to w' in $\mathcal{G}(C)$ whose threshold $M_\gamma=\mathrm{lcm}_{v\in\gamma}K(v)$ divides m. The minimal such M_γ form an antichain under divisibility (the connection thresholds *of the pair).*

(b) The swap graph depends only on $\mathcal{A}(m)$. If $m'=\mathrm{lcm}\{K(w):w\in\mathcal{A}(m)\}$ then $m'\mid m$ and $\mathcal{A}(m')=\mathcal{A}(m)$. Hence every swap-graph shape occurring for C occurs at a source in the finite lcm-closure of $\{K(w)\}$, and the classification of C is a finite computation.

*(c) (*monotonicity*) If $m\mid m^*$ then $\mathcal{A}(m)\subseteq\mathcal{A}(m^*)$; the classes at m^* are unions of classes at m, together with possibly new classes made of newly admissible arrangements. A class never splits, but $\nu(C,m)$ is not monotone in m.*

Proof. (a) A path has all its vertices admissible iff every $K(v)$, $v\in\gamma$, divides m, iff their lcm does; and a minimal set of thresholds is an antichain by definition. (b) m' divides m since each $K(w)$ with $w\in\mathcal{A}(m)$ does; $\mathcal{A}(m')\supseteq\mathcal{A}(m)$ since those $K(w)$ divide m', and $\mathcal{A}(m')\subseteq\mathcal{A}(m)$ because $m'\mid m$. (c) $K(w)\mid m\mid m^*$. For non-monotonicity see the examples below. □

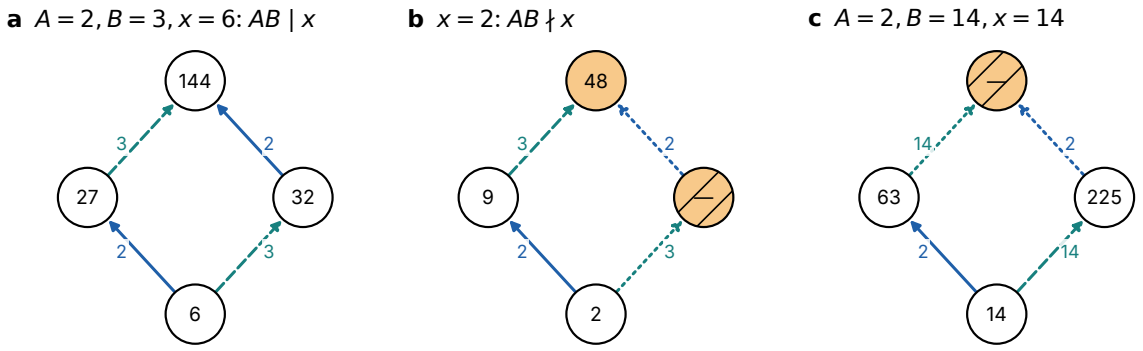


Figure 7: **Legal and forbidden swap diamonds (exact).** Each panel shows the states (circles) reached from a state x by the labels A (solid blue edges) and B (dashed teal edges); dotted edges and hatched orange nodes are inadmissible. (a) $A = 2$, $B = 3$, $x = 6$: $AB \mid x$, the diamond closes at 144. (b) $x = 2$: $A = 2$ is admissible but $B = 3$ is not, so only one side exists. (c) $A = 2$, $B = 14$, $x = 14$: both labels are admissible at 14 but $AB = 28 \nmid 14$, so after 2 the label 14 fails (63 is not divisible by 14) and after 14 the label 2 fails. Theorem illustrated: Proposition 4.5.

Source record: `D-T8` (`CW-0346`); status PROVED; treatment FULL.

Example 5.6 (class counts are not monotone)**.** For $C = \{2, 3, 14\}$: $\nu = 2$ at $m = 14$ and $\nu = 1$ at $m = 28$ (Theorem 5.8). For $C = \{2, 3, 8\}$ the thresholds are $K(2, 3, 8) = 2$, $K(3, 2, 8) = K(3, 8, 2) = 3$, $K(8, 3, 2) = 8$, $K(2, 8, 3) = K(8, 2, 3) = 16$ and one finds $\nu = 1, 2, 1$ at the sources $2, 8, 16$, which are totally ordered by divisibility: a class count can rise and fall along a divisibility chain.

5.3 Same content, different swap classes: the source-14 example

Example 5.7 (source 14)**.** From the source 14 the content $\{2, 3, 14\}$ has exactly two admissible arrangements (Example 4.8):

$$14 \xrightarrow{14} 225 \xrightarrow{3} 1200 \xrightarrow{2} 5400, \qquad 14 \xrightarrow{2} 63 \xrightarrow{3} 336 \xrightarrow{14} 5400.$$

They have the same source, the same content and the same endpoint 5400, and neither has a legal swap: exchanging $(14, 3)$ at the state 14 needs $42 \mid 14$; $(3, 2)$ at 225 needs $6 \mid 225$; $(2, 3)$ at 14 needs $6 \mid 14$; and $(3, 14)$ at 63 needs $42 \mid 63$ — all four fail by Proposition 4.5(c). The mechanism is resource contention: the prime 2 has capacity 1 at 14; both 2 and 14 consume it and only the label 3 (with $4 = 3 + 1$) produces it, so exactly one of $\{2, 14\}$ can run before 3. The two histories also have different fibre sequences and different occurrence events (Section 11).

The figure above is Figure 7; its caption states the exactness of what is drawn.

Theorem 5.8 (complete classification of $\{2, 3, 14\}$)**.** *The thresholds are* $K = 14$ *for* $(14, 3, 2), (2, 3, 14)$*,* $K = 28$ *for* $(14, 2, 3), (2, 14, 3)$*, and* $K = 21$ *for* $(3, 14, 2), (3, 2, 14)$*. The*

number of swap classes of $\{2,3,14\}$ *at the source* m *is*

$$\nu = \begin{cases} 0 & \text{if neither } 14 \text{ nor } 21 \text{ divides } m, \\ 2 & \text{if } 14 \mid m \text{ and } \gcd(m/14, 6) = 1, \\ 1 & \text{otherwise.} \end{cases}$$

In particular, the two arrangements $(14,3,2)$ *and* $(2,3,14)$ *lie in one class iff* $28 \mid m$ *or* $42 \mid m$ *(their connection thresholds), and from* $m = 14q$ *there are two classes iff* $\gcd(q,6) = 1$*. All admissible arrangements from* m *end at* $m \cdot 2700/7$*.*

Proof. The thresholds were computed in Example 4.8. Order the arrangements around the hexagon $\mathcal{G}(C)$ as $\sigma_0 = (14,3,2)$, $\sigma_1 = (3,14,2)$, $\sigma_2 = (3,2,14)$, $\sigma_3 = (2,3,14)$, $\sigma_4 = (2,14,3)$, $\sigma_5 = (14,2,3)$ (consecutive ones, and σ_5, σ_0, differ by an adjacent exchange). Then σ_0, σ_3 have threshold 14 and are antipodal, σ_1, σ_2 have threshold 21 and are adjacent, and σ_4, σ_5 have threshold 28 and are adjacent. Every threshold is a multiple of 7, so $7 \mid m$ is necessary for any arrangement to be admissible. Now $\mathcal{A}(m)$ is the union of the pairs whose threshold divides m. If $14 \nmid m$ then $28 \nmid m$ and $\mathcal{A}(m) \subseteq \{\sigma_1, \sigma_2\}$, which is empty if $21 \nmid m$ and one edge (one class) if $21 \mid m$. If $14 \mid m$ then $\sigma_0, \sigma_3 \in \mathcal{A}(m)$; adding the 28-pair when $28 \mid m$ gives the path $\sigma_3\sigma_4\sigma_5\sigma_0$, adding the 21-pair when $21 \mid m$ gives the path $\sigma_0\sigma_1\sigma_2\sigma_3$; in either case the graph is connected, while with neither it is two isolated vertices. Finally $14 \mid m$ and $21 \nmid m$ and $28 \nmid m$ means $14 \mid m$, $3 \nmid m$ and $m/14$ odd, i.e. $\gcd(m/14, 6) = 1$. The connection thresholds are $\operatorname{lcm}(14,28) = 28$ and $\operatorname{lcm}(14,21) = 42$. The common endpoint is $mR(2)R(3)R(14) = m \cdot \frac{9}{2} \cdot \frac{16}{3} \cdot \frac{225}{14} = m \cdot \frac{2700}{7}$. □

Source record: `D-T10` (`CW-0348`); status PROVED; treatment FULL.

Corollary 5.9 (an infinite family of complete histories)**.** *Let* $q \geq 1$ *be squarefree and coprime to* 42*. Then* $14q$ *is squarefree, and the two complete histories* $(14q; 14, 3, 2)$ *and* $(14q; 2, 3, 14)$*, both ending at* $5400q$*, lie in different swap classes.*

Proof. $14q$ is squarefree because q is squarefree and coprime to $2 \cdot 7$; $\gcd(q,6) = 1$ gives two classes by Theorem 5.8. □

Source record: `D-T5` (`CW-0342`); status PROVED; treatment FULL.
The figure above is Figure 8; its caption states the exactness of what is drawn.

Remark 5.10. Theorem 5.8 relaxes a restriction of the earliest release: there the pair was shown for q squarefree and coprime to 210; the condition on the prime 5 is unnecessary for the statement about fixed-source histories, as the proof shows (and $7 \nmid q$ is needed only for squarefreeness of the source).

6 Feeding relations, factorisation and components

Definition 6.1 (feeding digraph, blocks)**.** For labels $A \neq B$ say that A *feeds* B if $\gcd(A+1, B) > 1$ (a prime of $A+1$ divides B: running A produces a resource that B consumes). The *feeding digraph* of a multiset C has the distinct labels of C as vertices and the pairs (A, B) with A feeding B as edges. Its strongly connected components are the *blocks*; the condensation is acyclic, so the blocks can be listed in a *topological order*, feeders first: if A feeds B then the block of A is not later than the block of B.

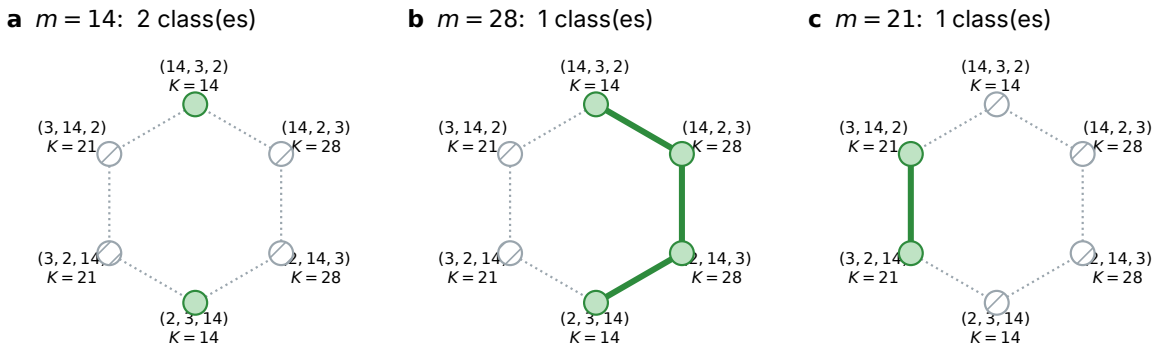


Figure 8: **The same content in different swap components (exact).** The hexagon $\mathcal{G}(\{2,3,14\})$ with the threshold K of each arrangement; admissible vertices at the source m are shaded green and joined by thick solid green lines, inadmissible ones are hatched. (a) $m = 14$: the antipodal pair $(14,3,2), (2,3,14)$, two classes, with the histories of Example 5.7. (b) $m = 28$: the 28-pair joins them, one class. (c) $m = 21$: only the 21-pair, one class. Theorem illustrated: Theorem 5.8.

Lemma 6.2 (NF: non-feeding pairs commute)**.** *If x does not feed y and (x, y) is admissible from a state s, then (y, x) is admissible from s. Consequently, for $x \neq y$ with x not feeding y, an adjacent pair (x, y) in an admissible word can always be exchanged to (y, x).*

Proof. $\gcd(x+1, y) = 1$ gives $\gcd(y, (x+1)^2) = 1$, so $K((x, y)) = xy$ by Proposition 4.5(a). Also $K((y, x)) = xy/\gcd(x, (y+1)^2)$ divides xy. Hence $K((y, x)) \mid K((x, y)) \mid s$. □

Source record: `NF` (`CW-0350`); status PROVED; treatment FULL.

Theorem 6.3 (block factorisation)**.** *Let C be a multiset (repeated labels allowed) with blocks $B_1, \ldots, B_r$ in topological order, C_i the sub-multiset of C on B_i, and $m_0 = m$, $m_i = m_{i-1} R(C_i)$. Then*

$$\nu(C, m) = \prod_{i=1}^{r} \nu(C_i, m_{i-1}),$$

where $\nu(C_i, y)$ is read as 0 if y is not an integer. In particular $\nu(C, m) = 0$ as soon as one factor is 0.

Proof. Sorting. Let w be admissible from m and let x, y be adjacent letters of w with x in a later block than y. Then x does not feed y (an edge $x \to y$ would put the block of x no later than that of y), so by Lemma 6.2 exchanging them is a legal swap. A stable bubble sort by block index reaches the block-sorted word $u_1 \cdots u_r$, where u_i is the subword of w on C_i, by legal swaps of letters of different blocks; so $u_1 \cdots u_r$ is admissible, hence each u_i is admissible from m_{i-1} (the state after $u_1 \cdots u_{i-1}$ is m_{i-1} whatever the orders inside the u_j). *Invariant.* Let $\pi_i(w)$ be the subword of w on C_i. A legal swap of two letters from different blocks leaves every π_i unchanged. A legal swap of two letters of the same block B_i changes only π_i, by one exchange of adjacent distinct letters (the letters are adjacent in w, hence in π_i); sorting w and the swapped word gives two admissible block-sorted words that differ by this exchange inside the i-th segment, so the exchange is a legal swap in $\mathrm{Sw}(C_i, m_{i-1})$. Therefore the tuple of classes

$([\pi_1(w)], \ldots, [\pi_r(w)])$ is constant on swap classes of C at m. *Bijection.* The map $w \mapsto ([\pi_i(w)])_i$ from swap classes to $\prod_i \{\text{classes of } \mathrm{Sw}(C_i, m_{i-1})\}$ is onto: concatenating admissible block words v_i from m_{i-1} gives an admissible word, because the entry state of block i is m_{i-1} whatever was chosen before. It is injective: two words with the same tuple are connected by sorting both and then moving inside each block along a path of $\mathrm{Sw}(C_i, m_{i-1})$; each such step is a legal swap of the whole word, since the state before block i is m_{i-1} in both. □

Source record: `F2` (`CW-0351`); status PROVED; treatment FULL.

Corollary 6.4 (acyclic feeding digraph)**.** *If every block is a single label (the feeding digraph is acyclic), then $\nu(C, m) \leq 1$ for every m. Moreover for the block-sorted word τ one has $K(\tau) \mid K(w)$ for every arrangement w, so $K(\tau)$ is the universal minimal threshold, and $\mathcal{A}(m)$ is a lower set of the weak order determined by τ (Section 7).*

Proof. Each C_i is a power of one letter and has one arrangement, so each factor of Theorem 6.3 is 0 or 1. For the second claim, a legal exchange of an inverted pair (relative to τ) turns an admissible word w into an admissible w' (Lemma 6.2); applying it at the source $m = K(w)$ shows $K(w') \mid K(w)$, and bubble sort ends at τ. □

Source record: `F2` (`CW-0351`); status PROVED; treatment FULL.

Remark 6.5. An acyclic feeding digraph is a *sufficient* condition for connectivity at every source; it is not necessary (a strongly connected content may never split; see Example 6.11). Conversely, a split needs a block with at least 3 labels: a block with at most two distinct labels, each once, has at most two arrangements, which are adjacent, so for contents of distinct labels such a block never splits.

6.1 Hub descent

Proposition 6.6 (hub-descent criterion)**.** *Call an arrangement h of C a* hub *if $K(h) \mid K(w)$ for all arrangements w and every w has a path $w = v_0, v_1, \ldots, v_s = h$ in $\mathcal{G}(C)$ with $K(v_{i+1}) \mid K(v_i)$. If a hub exists then $\nu(C, m) \leq 1$ for all m.*

Proof. If $w \in \mathcal{A}(m)$ then $K(h) \mid K(w) \mid m$ and every v_i on the descent path has $K(v_i) \mid K(w) \mid m$, so w is joined to h inside $\mathrm{Sw}(C, m)$. □

Source record: `Hub-descent` (`CW-0352`); status PROVED; treatment FULL.

The criterion is sufficient but not necessary. *Finite-exact census:* over the 4060 contents of three distinct labels in $\{1, \ldots, 30\}$, 3408 have a hub and never split, none has a hub and a split, 525 have neither a hub nor a split, and 127 split (all without a hub); for four distinct labels in $\{1, \ldots, 14\}$ the counts are 773/0/125/103.

Source record: `CONNECTIVITY:Census-on-three-labels-FINITE` (`CW-0353`); status FINITE-EXACT; treatment FINITE.

6.2 Three distinct labels

For a multiset S of labels, $\kappa(S)$ is as in Section 4. For three distinct labels the arrangement graph is the hexagon.

Theorem 6.7 (exact classification of three distinct labels)**.** *Let $C = \{x, y, z\}$ consist of three distinct labels. The swap graph of C at m is disconnected if and only if there is a labelling (a, b, c) of C such that*

$$a \mid m, \quad c \mid m, \quad \kappa\{a, b\} \mid m, \quad \kappa\{b, c\} \mid m, \quad \kappa\{a, b, c\} \mid m, \qquad b \nmid m, \quad \kappa\{a, c\} \nmid m.$$

In that case $\mathcal{A}(m) = \{(a, b, c), (c, b, a)\}$, an antipodal pair of the hexagon, and $\nu(C, m) = 2$. A content has a split source iff for some labelling $L = \operatorname{lcm}(a, c, \kappa\{a, b\}, \kappa\{b, c\}, \kappa\{a, b, c\})$ is divisible by neither b nor $\kappa\{a, c\}$; the split sources of that labelling are the multiples m of L with $b \nmid m$ and $\kappa\{a, c\} \nmid m$.

Proof. ($\Leftarrow$) By Corollary 4.3 the prefix multisets of (a, b, c) are $\{a\}, \{a, b\}, C$ and of (c, b, a) are $\{c\}, \{c, b\}, C$, all feasible by hypothesis ($\kappa\{a\} = a$, $\kappa\{c\} = c$). The words (a, c, b) and (c, a, b) have the prefix multiset $\{a, c\}$ and (b, a, c), (b, c, a) have $\{b\}$; both are infeasible by hypothesis ($\kappa\{b\} = b$). So $\mathcal{A}(m)$ is the antipodal pair, which is disconnected. ($\Rightarrow$) Let $\mathcal{A}(m)$ be disconnected and u, w in different components. If u, w were at distance 2 in the hexagon, their unique common neighbour v arises from u and from w by exchanges at different positions, so each prefix multiset of v is a prefix multiset of u or of w; by Corollary 4.3 $K(v) \mid \operatorname{lcm}(K(u), K(w)) \mid m$, so $v \in \mathcal{A}(m)$ and u, w would be connected. Hence any two vertices in different components are antipodal. A vertex has a unique antipode, so no component contains two vertices (they would both be antipodal to every vertex of another component) and there are at most two components. Thus $\mathcal{A}(m) = \{w, w^{\mathrm{rev}}\}$; write $w = (a, b, c)$. Its prefix multisets $\{a\}, \{a, b\}, \{c\}, \{c, b\}, C$ are feasible, which gives the five divisibilities; the four other arrangements are inadmissible, and (a, c, b) (resp. (b, a, c)) has $\{a, c\}$ (resp. $\{b\}$) as its only prefix multiset not among the feasible ones, so $\kappa\{a, c\} \nmid m$ and $b \nmid m$. The last statements follow by taking the least m satisfying the divisibilities. □

Source record: `T3` (`CW-0354`); status PROVED; treatment FULL.

Remark 6.8. The mechanism in words: b alone is infeasible (it consumes something that only a or c produces) and $\{a, c\}$ together is infeasible (they consume something only b produces); everything else is feasible. The two histories can only be run "through b" from the ends and cannot be exchanged into each other.

6.3 The family $\{2, 3, N\}$

Theorem 6.9 (the family $\{2, 3, N\}$)**.** *Let $N \geq 4$, $N \neq 2, 3$, and $C = \{2, 3, N\}$.*

(a) The feeding digraph of C is strongly connected iff $N \equiv 2$ or $3 \pmod 6$ (then $N \geq 8$). Otherwise its blocks are $\{2, 3\}$ and $\{N\}$ and $\nu(C, m) \leq 1$ for every m.

(b) If $N \equiv 2, 3 \pmod 6$ then $\nu(C, m) = 2$ iff $m = Nt$ with $\gcd(t, 6) = 1$, and $\nu(C, m) \leq 1$ otherwise. In particular C has a split source iff $N \equiv 2, 3 \pmod 6$, and then the minimal split source is N itself, where $\mathcal{A}(N)$ is $\{(2, 3, N), (N, 3, 2)\}$ for $N \equiv 2$ and $\{(N, 2, 3), (3, 2, N)\}$ for $N \equiv 3 \pmod 6$.

Proof. (a) 2 and 3 feed each other ($\gcd(3, 3) = 3$, $\gcd(4, 2) = 2$). N feeds 2 iff N is odd; N feeds 3 iff $3 \mid N + 1$; 2 feeds N iff $3 \mid N$; 3 feeds N iff N is even. So N lies in the block of $\{2, 3\}$ iff (N odd or $N \equiv 2 \bmod 3$) and (N even or $3 \mid N$), which is $N \equiv 2, 3 \pmod 6$ by inspecting the six residues. Otherwise the blocks $\{2, 3\}$ and $\{N\}$ each have at most one

arrangement up to an adjacent exchange, so every factor of Theorem 6.3 is ≤ 1. (b) Let $N \equiv 2 \pmod 6$: N is even, $3 \nmid N$, $3 \mid N+1$; write $a = v_2(N) \ge 1$, $N = 2^a R$ with R odd and prime to 3, and $c = v_3(N+1) \ge 1$. Direct computation of Δ gives $\kappa\{2\} = 2$, $\kappa\{3\} = 3$, $\kappa\{N\} = N$, $\kappa\{2,3\} = 1$ (as $R(2)R(3) = 24$), $\kappa\{2,N\} = 2^{a+1}R = 2N$, $\kappa\{3,N\} = 2^{\max(0,a-4)}R$ and $\kappa\{2,3,N\} = 2^{\max(0,a-3)}R$ (for the primes $q \ge 5$ only the consumption $v_q(N)$ matters, and 3 is produced by 2, N and consumed only by 3). By Corollary 4.3, $K(2,3,N) = 2^{\max(1,a-3)}R$ and $K(N,3,2) = N$ both divide N, while $K(2,N,3)$, $K(N,2,3)$ are multiples of $2N$ and $K(3,\cdot,\cdot)$ of 3, none of which divides N. So $\mathcal{A}(N) = \{(2,3,N),(N,3,2)\}$, antipodal, $\nu(C,N) = 2$. For general m use Theorem 6.7: the middle label b must not divide m while the two ends do. If $b = 2$ then $2 \nmid m$, but N is an end, so $2 \mid N \mid m$, a contradiction. If $b = N$ then one of $\kappa\{a,b\}, \kappa\{b,c\}$ is $\kappa\{2,N\} = 2N$ and must divide m, forcing $N \mid m$, a contradiction. So $b = 3$, the ends are $2, N$, and the conditions of Theorem 6.7 read: $N \mid m$ (all of $\kappa\{3,N\}, \kappa\{2,3,N\}$ divide N), $3 \nmid m$ and $2N \nmid m$; with $m = Nt$ these say $3 \nmid t$ and t odd. For $N \equiv 3 \pmod 6$ the roles of 2 and 3 are exchanged: N is odd, $N = 3^b R$ with $\gcd(R,6) = 1$, $b \ge 1$, $c = v_2(N+1) \ge 1$, $\kappa\{3,N\} = 3^{b+1}R = 3N$, $\kappa\{2,N\} = 3^{\max(0,b-2)}R$, $\kappa\{2,3,N\} = 3^{\max(0,b-1)}R$, $\kappa\{2,3\} = 1$; $\mathcal{A}(N) = \{(N,2,3),(3,2,N)\}$, and the same argument forces the middle label to be 2 and gives $\nu(C,m) = 2$ iff $N \mid m$, $2 \nmid m$, $3N \nmid m$, i.e. $m = Nt$, $\gcd(t,6) = 1$. In both cases $N \ge 8$ is automatic since $N \ge 4$ and $N \notin \{2,3\}$ and $N \equiv 2,3 \pmod 6$. □

Remark 6.10 (status relative to the source). The source release proves the arithmetic of the family and records only a finite-exact range ($4 \le N \le 400$) for the converse. Part (a) together with Theorem 6.3 and the argument of (b) prove the converse for all $N \ge 4$; the finite range is corroboration.

Source record: `D-T9` (`CW-0347`); status MIXED; treatment SPLIT.

Source record: `OBSTRUCTIONS:Parameter-family-2-3-N-PROV` (`CW-0355`); status PROVED; treatment FULL. The figure above is Figure 9; its caption states the exactness of what is drawn. The figure above is Figure 10; its caption states the exactness of what is drawn.

Example 6.11 (strongly connected, never split)**.** The contents $\{2,4,15\}$, $\{2,5,9\}$ and $\{2,6,7\}$ have strongly connected feeding digraphs and $\nu \le 1$ at every source in the lcm-closure of their thresholds. Strong connectivity of the feeding digraph is therefore necessary but not sufficient for a split.

Proposition 6.12 (finite-exact: all three-label splits are antipodal)**.** *Among the* $\binom{30}{3} = 4060$ *contents of three distinct labels in* $\{1,\dots,30\}$*, exactly* 257 *have a strongly connected feeding digraph and exactly* 127 *have a split source; each split content is strongly connected, so* 130 *strongly connected contents never split. For each of the* 4060 *contents and every source in the lcm-closure of its thresholds, the criterion of Theorem 6.7 agrees with the direct class count (no violation).* FINITE-EXACT.

Source record: `THEOREMS_V002:Census-FINITE-EXACT-results` (`CW-0349`); status FINITE-EXACT; treatment FINITE.

7 Feasible-subset geometry and topological obstructions

Throughout this section C is a set of $k \ge 3$ *distinct* labels (repeated labels need a different construction, Problem 8.11).

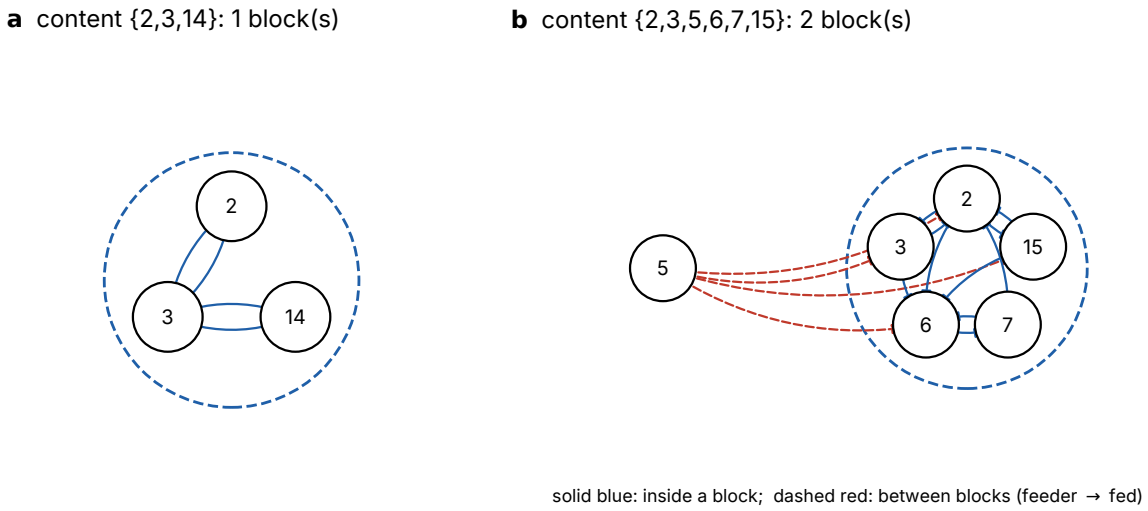


Figure 9: **Feeding digraphs and blocks (exact).** Dashed blue circles are strongly connected blocks; solid blue arrows are feeding edges inside a block and dashed red arrows are feeding edges between blocks (feeder to fed). (a) $\{2, 3, 14\}$: one block (3 feeds 14 and 14 feeds 3; 14 does not feed 2 or conversely). (b) $\{2, 3, 5, 6, 7, 15\}$: the label 5 is a block of its own feeding the block $\{2, 3, 6, 7, 15\}$. Theorems illustrated: Theorem 6.3, Corollary 6.4.

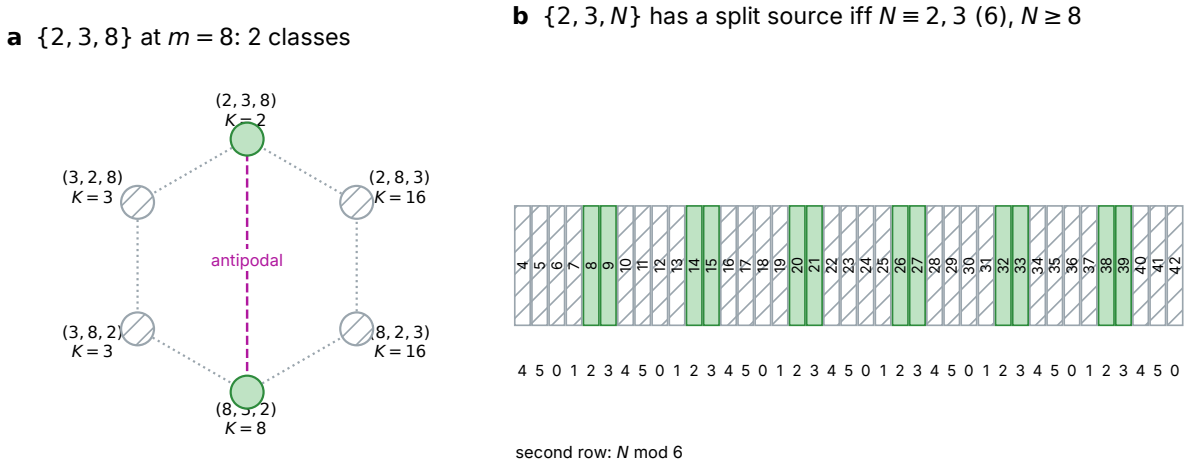


Figure 10: **Three-label split and the antipodal orderings (exact; the strip is exact in the range shown).** (a) The hexagon of $\{2, 3, 8\}$ at $m = 8$ with the threshold K of each arrangement: only the antipodal pair $(2, 3, 8), (8, 3, 2)$ is admissible (two classes). (b) The integers $N = 4, \ldots, 42$ with residues mod 6: shaded cells are those for which $\{2, 3, N\}$ has a split source, namely $N \equiv 2, 3 \pmod 6$ with $N \geq 8$. Theorems illustrated: Theorem 6.7, Theorem 6.9.

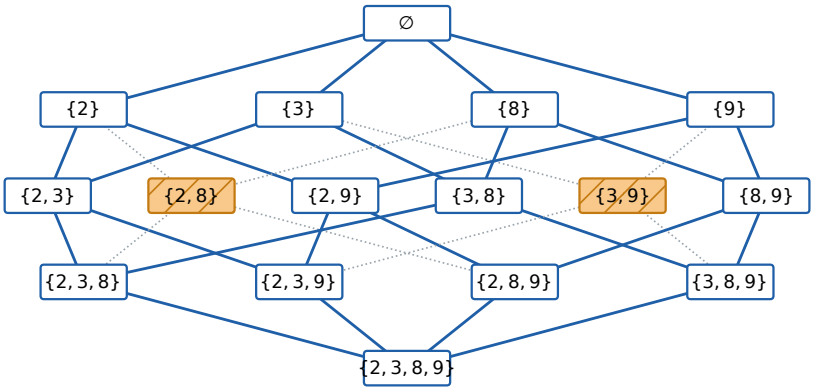


Figure 11: **Feasible-subset diagram of $\{2,3,8,9\}$ at $m = 72$ (exact).** The Boolean lattice of subsets; the two infeasible proper subsets $\{2,8\}$ and $\{3,9\}$ are hatched orange, and the edges at them are dotted grey. The maximal chains $\emptyset \subset \cdots \subset C$ that avoid them are the admissible arrangements: 16 of the 24. Theorem illustrated: Corollary 4.3, Theorem 7.5.

7.1 Feasible subsets and maximal chains

By Corollary 4.3 an arrangement w is admissible from m iff every prefix set of w is feasible at m. For distinct labels the admissible histories of content C are therefore exactly the *maximal chains* $\emptyset \subset S_1 \subset \cdots \subset S_k = C$ of feasible subsets (S_j the set of the first j letters). A set $S \subseteq C$ is *infeasible* at m iff for some prime p,

$$\sum_{x \in S} \Delta_p(x) < -v_p(m),$$

a knapsack-type threshold: S consumes more of the resource p than m supplies plus what its members themselves produce. The whole route theory is a theory of the family of infeasible proper subsets (Figure 11). The set C itself is feasible whenever $\mathcal{A}(m) \neq \emptyset$.

7.2 The route complexes

The permutohedron P_k is the convex hull of the points $(\sigma(1), \ldots, \sigma(k)) \in \mathbb{R}^k$, σ a permutation; it is a simple $(k-1)$-polytope whose boundary is homeomorphic to S^{k-2}, and its faces correspond to the ordered set partitions $(B_1|\cdots|B_j)$ of the labels, with dimension $k-j$ and vertex set the arrangements that list B_1 first (in any order), then $B_2, \ldots$ (see Ziegler [15, Example 0.10]). The vertices are the arrangements and the edges are the exchanges of adjacent letters.

Definition 7.1 (route complexes)**.** For a set $\mathcal{A}$ of arrangements, the *permutohedral route complex* $K_{\mathcal{A}}$ is the full subcomplex of the face complex of P_k on the vertex set $\mathcal{A}$: a face is kept iff all its vertices lie in $\mathcal{A}$. The *cubical route complex* $Q_{\mathcal{A}}$ is the subcomplex of faces all of whose blocks have size ≤ 2 (products of edges): its vertices are the admissible histories, its edges the legal swaps, and its squares and cubes the sets of independent swaps (Lemma 8.1). We write $K_m = K_{\mathcal{A}(m)}$ and Q_m. Homology is over $\mathbb{F}_2$ unless stated. The 1-skeleton of either complex is

the swap graph, so β_0 counts swap classes. A hexagon (braid relation $xyz \leftrightarrow zyx$) is a 2-face of K_m iff all six arrangements of the three letters are admissible, and is *never* a cell of Q_m.

Remark 7.2. Faces of P_k containing a given vertex v correspond to subsets of the $k-1$ adjacent position pairs of v (the blocks are the maximal intervals joined by the chosen pairs); this is the simplicity of P_k and is used in Theorem 7.4.

7.3 Alexander duality

Write $\mathcal{R} = \mathfrak{S}_k \setminus \mathcal{A}$ for the set of non-admissible arrangements and $\mathcal{N}(\mathcal{R})$ for the simplicial complex on $\mathcal{R}$ whose simplices are the subsets of $\mathcal{R}$ contained in a common proper face of P_k. The nerve theorem and Alexander duality used below are in [14]; the permutohedron is described in [15].

Theorem 7.3 (duality for route complexes)**.** *Let $\emptyset \neq \mathcal{A} \neq \mathfrak{S}_k$. Then $K_{\mathcal{A}} \subset \partial P_k \cong S^{k-2}$ and, over $\mathbb{F}_2$, for all i*

$$\dim \tilde{H}_i(K_{\mathcal{A}}) = \dim \tilde{H}^{\,k-3-i}\big(\mathcal{N}(\mathcal{R})\big).$$

Two vertices of $\mathcal{R}$ lie in a common proper face iff they share a proper prefix set; hence $\mathcal{N}(\mathcal{R})$ has the connected components of the graph on $\mathcal{R}$ joined by "share a proper prefix set". Consequently $\tilde{\beta}_{k-2}(K_{\mathcal{A}}) = 0$ and $\tilde{\beta}_{k-3}(K_{\mathcal{A}}) = c(\mathcal{R}) - 1$, where $c(\mathcal{R})$ is the number of such components. In particular for $k = 3$, $\tilde{\beta}_1 = 0$; for $k = 4$, $\beta_2 = 0$ and $\beta_1 = c(\mathcal{R}) - 1$; for $k = 5$, $\beta_3 = 0$ and $\beta_2 = c(\mathcal{R}) - 1$.

Proof. $K_{\mathcal{A}}$ is a proper nonempty subcomplex of the sphere ∂P_k (a vertex of $\mathcal{R}$ lies outside it). A point $x \in \partial P_k$ lies in the relative interior of a unique face F, and $x \in K_{\mathcal{A}}$ iff all vertices of F lie in $\mathcal{A}$. Hence the complement of $K_{\mathcal{A}}$ is the union, over $v \in \mathcal{R}$, of the open stars $\mathrm{st}(v)$ (the union of the relative interiors of the faces containing v). Each open star is open in ∂P_k (its complement is the union of the faces not containing v) and contractible (it is star-shaped from v). For a family $v_1, \ldots, v_n$ the intersection $\bigcap \mathrm{st}(v_i)$ is the union of the relative interiors of the faces containing all v_i; these are the faces containing the smallest face Φ containing all v_i (an intersection of faces is a face), so the intersection is $\mathrm{st}(\Phi)$, which is contractible, provided Φ is a proper face, and it is empty otherwise (there is no proper face containing all v_i). By the nerve theorem for open covers [14, Cor. 4G.3] the complement of $K_{\mathcal{A}}$ is homotopy equivalent to the nerve of this cover, which is $\mathcal{N}(\mathcal{R})$. Alexander duality [14, Prop. 3.46] applied to the compact, locally contractible, nonempty proper subspace $K_{\mathcal{A}}$ of S^{k-2} gives $\tilde{H}_j(S^{k-2} \setminus K_{\mathcal{A}}) \cong \tilde{H}^{k-3-j}(K_{\mathcal{A}})$; over a field $\tilde{H}^q$ and $\tilde{H}_q$ of a finite complex have equal dimension, which gives the displayed equality after exchanging the roles of the two sides. Two vertices lie in a common proper face iff they lie in a face $(B_1|\cdots|B_j)$ with $j \geq 2$, i.e. iff they both begin with the proper subset B_1, i.e. share a proper prefix set. Finally $\tilde{H}^{-1}$ of a nonempty complex vanishes (this gives $\tilde{\beta}_{k-2} = 0$) and $\tilde{\beta}^0$ counts components minus one. □

Source record: `T-DUAL` (`CW-0359`); status PROVED; treatment FULL.

Check. The identity $\beta_{k-3} = c(\mathcal{R}) - 1$ (and $\beta_{k-2} = 0$) was verified against independent direct homology computations for 1772 complexes with $k = 4, 5$ (a sample of contents with labels ≤ 14, resp. ≤ 11, and all sources in the lcm-closure), with no violation (`results/RESULTS_II_part3.json`). FINITE-EXACT corroboration.

7.4 Lower sets are contractible

Fix a total order τ on the labels. The *weak order* on arrangements is $u \le v$ iff the inversion set of u (pairs ordered differently from τ) is contained in that of v; its minimum is the τ-sorted word b and its maximum the reverse. A set $\mathcal{A}$ is a *lower set* if it is closed under replacing an adjacent inverted pair by the non-inverted one.

Theorem 7.4 (lower sets)**.** *If $\mathcal{A}$ is a nonempty lower set of the weak order, then $K_{\mathcal{A}}$ is contractible. Consequently, if the feeding digraph of C is acyclic, then K_m is empty or contractible for every source m.*

Proof. If $\mathcal{A} = \mathfrak{S}_k$ then $K_{\mathcal{A}}$ is the single cell P_k; otherwise $b \in \mathcal{A}$ since $\mathcal{A}$ is nonempty and a lower set. Induct on $|\mathcal{A}|$; for $\mathcal{A} = \{b\}$ the complex is a point. Let $v \neq b$ be maximal in $\mathcal{A}$. Faces of P_k through v correspond to subsets J of the $k-1$ adjacent position pairs of v. If J contains an ascent of v (an adjacent pair in τ-order), the face contains the neighbour of v obtained by exchanging it, which has one more inversion and is not in $\mathcal{A}$ by maximality. If J consists of descents only, each block of the face is a segment of a decreasing run, so all vertices of the face are obtained from v by sorting inside blocks and are $\le v$, hence in $\mathcal{A}$. So the faces of $K_{\mathcal{A}}$ through v correspond to the subsets of $\mathrm{Desc}(v)$, nonempty because $v \neq b$: the link of v in $K_{\mathcal{A}}$ is a full simplex L (contractible), and the closed star of v is a cone over it. Then $K_{\mathcal{A}} = K_{\mathcal{A}\setminus\{v\}} \cup \mathrm{star}(v)$ with $K_{\mathcal{A}\setminus\{v\}} \cap \mathrm{star}(v) = L$; gluing a cone along its contractible base does not change the homotopy type, so $K_{\mathcal{A}} \simeq K_{\mathcal{A}\setminus\{v\}}$, and $\mathcal{A} \setminus \{v\}$ is again a nonempty lower set. For the consequence: choose τ so that feeders come first; if x does not feed y and x is placed after y in τ-order... precisely, an inverted adjacent pair (x, y) (x after y in τ) has x not feeding y because all feeding edges go forward in τ; by Lemma 6.2 replacing it by (y, x) keeps an admissible word admissible. So $\mathcal{A}(m)$ is a lower set. □

Source record: `T-LOWER` (`CW-0360`); status PROVED; treatment FULL.

Check. For 303 nonempty route complexes of contents of four distinct labels with acyclic feeding digraph (labels ≤ 15, sources among the thresholds), every K_m has Betti vector $(1, 0, 0, 0)$; no violation. FINITE-EXACT corroboration.

7.5 Complementary pairs and spheres

Theorem 7.5 (complementary-pair deadlock)**.** *Let $C = E \sqcup O$ be a partition into nonempty sets and suppose the infeasible proper subsets of C at m are exactly E and O. Then K_m has the $\mathbb{F}_2$-homology of the sphere S^{k-3}: $\beta_0 = 1$ and $\beta_{k-3} = 1$ with all other Betti numbers zero (for $k = 3$: two points, $\beta_0 = 2$, the antipodal pair of Theorem 6.7).*

Proof. The non-admissible arrangements are those having E or O as a prefix set, i.e. $\mathcal{R} = F_{E|O} \sqcup F_{O|E}$, the vertex sets of two facets. A vertex v of the first facet and a vertex w of the second have no common proper prefix set: a proper prefix set of v of size $s \le |E|$ is a subset of E and for $s \ge |E|$ contains E, symmetrically for w with O; equality of such sets is impossible unless the set is C (the sets would have to lie in both E and O, or contain both). Hence $\mathcal{N}(\mathcal{R})$ has exactly two components, each a full simplex (all vertices of a facet lie in a common proper face). So $\tilde{H}^0(\mathcal{N}(\mathcal{R}))$ has rank one and all other reduced groups vanish, and Theorem 7.3 gives $\tilde{\beta}_{k-3} = 1$ and all other reduced Betti numbers 0. □

Source record: `T-SPHERE` (`CW-0361`); status PROVED; treatment FULL.

Remark 7.6 (scope). The homotopy type of K_m is not determined by this argument: it is a statement about $\mathbb{F}_2$-homology. (That K_m is homotopy equivalent to S^{k-3} would follow if the union of the open stars of the vertices of each facet were an open ball in the sphere, which is plausible but not proved here.) The arithmetic hypothesis (infeasible subsets exactly E and O) is a condition on (C, m) that is checked case by case, not a theorem.

Proposition 7.7 (arithmetic realisation; finite-exact)**.** *Let X be a set of labels $n \equiv 2, 3 \pmod 6$, $n \geq 8$, $C = \{2, 3\} \cup X$ and $m = \prod_{x \in X} x$. For all $X \subseteq \{8, 9, 14, 15, 20, 21\}$ with $1 \leq |X| \leq 3$ (41 contents) the infeasible proper subsets of C at m are exactly $E = \{x \in C : x \equiv 2 \bmod 6\}$ and $O = \{x \in C : x \equiv 3 \bmod 6\}$, and the Betti vector of K_m is that of S^{k-3}.* FINITE-EXACT.

Two instances are the examples that drive the manuscript.

Example 7.8 (loop: $\{2, 3, 8, 9\}$ at 72)**.** $E = \{2, 8\}$, $O = \{3, 9\}$, $m = 8 \cdot 9 = 72$. Of the 24 arrangements 16 are admissible, the swap graph is connected, and K_{72} has f-vector $(16, 20, 4)$ (four 2-faces) and Betti vector $(1, 1, 0, 0)$: a loop of legal swaps that is not filled. The cubical complex Q_{72} has the same f-vector and the same Betti vector. The loop is invisible to the connectivity classification (Figure 11, Figure 12).

Example 7.9 (void: $\{2, 3, 8, 9, 14\}$ at 1008)**.** $E = \{2, 8, 14\}$, $O = \{3, 9\}$, $m = 8 \cdot 9 \cdot 14 = 1008$. There are 96 admissible arrangements out of 120; K_{1008} has f-vector $(96, 180, 98, 12)$ and Betti vector $(1, 0, 1, 0, 0)$: a 2-dimensional void. The cubical complex Q_{1008} has f-vector $(96, 180, 66)$ and Betti vector $(1, 19, 0, 0, 0)$: nineteen loops and no void, so the genuine obstruction is seen only after the braid hexagons are filled (Figure 13).

Proposition 7.10 (cubical versus permutohedral; finite-exact)**.** *For the full sets of arrangements $(P(C) \mid m)$ of $k = 3, 4, 5$ distinct labels, K_m is a single cell (contractible), while Q_m has Betti vectors $(1, 1)$, $(1, 7)$, $(1, 31)$ respectively, computed here for $C = \{2, 3, 5\}$, $\{2, 3, 5, 7\}$, $\{2, 3, 5, 7, 11\}$ at $m = P(C)$. So a cube-only complex reports loops that are mere braid relations. For $\{2, 3, 8, 9\}$ at 72 the two complexes agree (Example 7.8); for $\{2, 3, 8, 9, 14\}$ at 1008 they differ in the way stated in Example 7.9.*

Source record: `ROUTE_TOPOLOGY:Cubical-versus-permutohedral` (`CW-0362`); status FINITE-EXACT; treatment FINITE.

7.6 Loop family and census

Proposition 7.11 (loop family; finite-exact)**.** *For $8 \leq N < M < 40$ with $N, M \equiv 2, 3 \pmod 6$, the content $\{2, 3, N, M\}$ has $\beta_1(K_{NM}) > 0$ (computed here for all such pairs). The programme's records state the converse over $4 \leq N < M \leq 60$ (a first Betti number appears at some source only if $N, M \geq 8$ and $N, M \equiv 2, 3 \bmod 6$); this converse was not re-run here and is* CITED.

Proposition 7.12 (census; cited)**.** *The programme's census over all sources in the lcm-closure records, for $k = 3$ (labels ≤ 30): 16 378 (content, source) pairs, Betti $(1, 0, 0)$ for 16 251 and $(2, 0, 0)$ for 127; for $k = 4$ (labels ≤ 18): 27 751 pairs, of which 27 321 contractible, 16 with $(1, 1, 0, 0)$, 405 with $(2, 0, 0, 0)$, 7 with $(3, 0, 0, 0)$, 2 with $(4, 0, 0, 0)$; for $k = 5$ (labels ≤ 11): 8 884 pairs, 8 617 contractible, 21 with $(1, 1, 0, 0, 0)$, 235 with $(2, 0, 0, 0, 0)$, 11 with $(3, 0, 0, 0, 0)$. No $\beta_i > 0$ for $i \geq 2$ occurs outside the void of Example 7.9 in these ranges. These are inherited finite computations,* CITED, *not re-run here.*

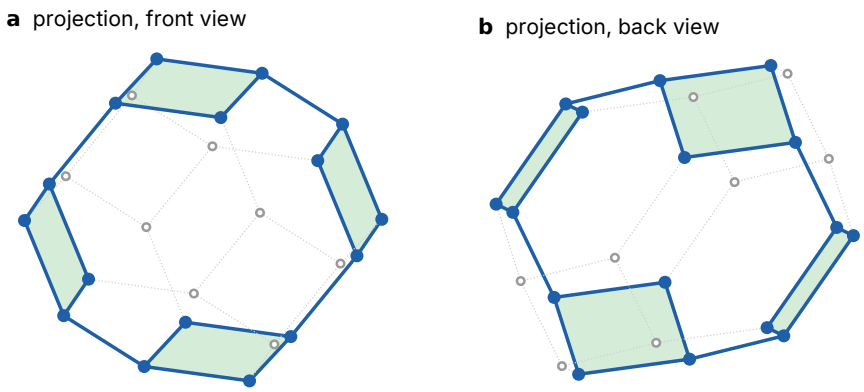


Figure 12: **The permutohedral route complex of $\{2,3,8,9\}$ at $m = 72$ (exact; projection of a 3-dimensional polytope to the plane, front and back views).** Filled blue dots are admissible arrangements, hollow dots are inadmissible; solid blue lines are legal swaps and dotted grey lines are non-legal exchanges; green faces are the four 2-faces (squares and hexagons) all of whose vertices are admissible. The unfilled cycle of legal swaps is the generator of $H_1 = \mathbb{F}_2$. The picture is a projection; the object is the 3-dimensional permutohedron's boundary S^2. Theorems illustrated: Theorem 7.5, Example 7.8.

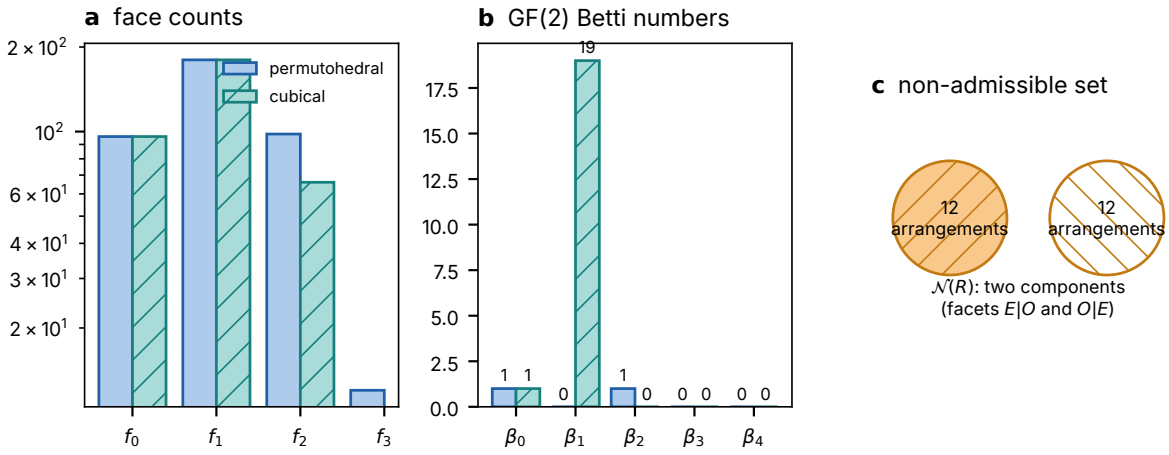


Figure 13: **The void of $\{2,3,8,9,14\}$ at $m = 1008$ (exact; incidence data, not a drawing of the 4-dimensional polytope).** (a) Face counts of the permutohedral complex K_{1008} and of the cubical complex Q_{1008} ($f_0, \ldots, f_3$; logarithmic scale). (b) $\mathbb{F}_2$-Betti numbers: $(1,0,1,0,0)$ for K_{1008} (blue, plain) and $(1,19,0,0,0)$ for Q_{1008} (teal, hatched). (c) The non-admissible set $\mathcal{R}$ has 24 arrangements in two components of $\mathcal{N}(\mathcal{R})$ (12 and 12, the facets $E|O$ and $O|E$), whence $\beta_2 = 2 - 1 = 1$ by Theorem 7.3. Theorems illustrated: Theorem 7.3, Theorem 7.5, Example 7.9.

Source record: ROUTE_TOPOLOGY:Census-FINITE-EXACT-contents (CW-0363); status FINITE-EXACT; treatment FINITE.

Problem 7.13 (open: wedge of spheres; realisable types)**.** Is K_m always a wedge of spheres (every complex found has the homology of one)? Which families of infeasible subsets occur as the infeasible family of an arithmetic content? Is every sphere S^j, $0 \le j \le k-3$, realised? Repeated labels need a construction by chains of sub-multisets: Problem 8.11.

Source record: ROUTE_TOPOLOGY:Statement-of-the-general-conje (CW-0364); status OPEN; treatment OPEN.

8 Higher-dimensional history complexes

8.1 Commuting swaps and route cubes

Lemma 8.1 (Independence Lemma)**.** *Let h be a history and $i_1 < \cdots < i_d$ positions with $i_{k+1} \ge i_k + 2$ such that the pair (N_{i_k}, N_{i_k+1}) consists of distinct labels whose exchange is legal in h. Then for every $b \in \{0,1\}^d$ the history h_b obtained by performing the chosen exchanges is admissible and has the same endpoint, and the h_b are pairwise distinct. They are the vertices of a d-cube whose edges are single legal swaps.*

Proof. Induct on the blocks left to right. Both orders of the k-th pair lead to the same state, so the state before the $(k+1)$-st pair does not depend on the choices already made; legality of the $(k+1)$-st exchange is a statement about that state only (Proposition 4.5(c)) and holds in every variant. All other letters are unchanged. The words are pairwise distinct since they differ in at least one of the chosen pairs. □

Source record: D-T3 (CW-0340); status PROVED; treatment FULL.

Theorem 8.2 (route cubes of 72^r)**.** *For every $r \ge 1$ the complete histories ending at 72^r contain an r-cube: from the source 1, the prefix 1^r followed by r blocks, each independently $(1,2)$ or $(2,1)$.*

Proof. After the prefix the state is 4^r; block j starts at $4^r\,18^{j-1}$, which is even, so $AB = 2$ divides it and the exchange $(1,2) \leftrightarrow (2,1)$ is legal by Proposition 4.5(c). Each block multiplies by $R(1)R(2) = 18$, so all 2^r histories end at $4^r 18^r = 72^r$, and Lemma 8.1 applies. □

Proposition 8.3 (the cube dimension of 72^r is not r; finite-exact)**.** *Let $\delta(r)$ be the maximum cube dimension (the largest number of pairwise non-overlapping legal swaps in one history) over the complete histories of 72^r. Then $\delta(1), \ldots, \delta(4) = 1, 2, 4, 5$; trivially $\delta(r) \le \lfloor l/2 \rfloor$ for a history of length l. An explicit 4-cube in 72^3: source 3, word $(3,1,2,1,2,1,2,1)$, legal exchanges at the position pairs $(1,2),(3,4),(5,6),(7,8)$, endpoint $3 \cdot \frac{16}{3} \cdot 4^4 \cdot (\frac{9}{2})^3 = 72^3$.* FINITE-EXACT *for $r \le 4$.*

Remark 8.4 (local does not imply global at overlaps). Lemma 8.1 concerns non-overlapping exchanges. For overlapping exchanges the braid hexagon of three distinct labels N_1, N_2, N_3 needs all six arrangements, which by Proposition 4.6 happens iff $N_1N_2N_3 \mid m$. The history $(2; 1, 2, 4)$ $(2 \to 8 \to 36 \to 225)$ has both adjacent exchanges legal ($(2,1,4)$ and $(1,4,2)$ are admissible) but only 3 of the 6 arrangements are admissible, since $8 \nmid 2$; and for $m = 60$ the labels $2, 6, 10$ are pairwise exchangeable $(12, 20, 60 \mid 60)$ while $120 \nmid 60$.

r	histories	swap classes	$(m_0, \text{content})$ groups	max cube dim.	$\beta_*(Q)$ $(\mathbb{F}_2)$
1	4	3	3	1	(3)
2	100	22	22	2	(22, 7)
3	3 601	142	141	4	(142, 368, 3)
4	194 864	1 114	—	5	—

Table 2: **Swap-class structure of the complete histories of 72^r (finite-exact).** Classes are counted over all (squarefree) sources. Q is the cubical route complex whose vertices are all complete histories of 72^r and whose cells are sets of independent legal swaps; its f-vectors are $(4, 1)$, $(100, 96, 11)$ and $(3601, 6366, 2832, 294, 4)$ (Euler characteristics 3, 15, -223 agree with the Betti vectors). The class counts for $r \leq 3$ and the cube dimensions for $r \leq 4$ agree with the programme's records; the value 1 114 for $r = 4$ and the complex Q were computed in this work.

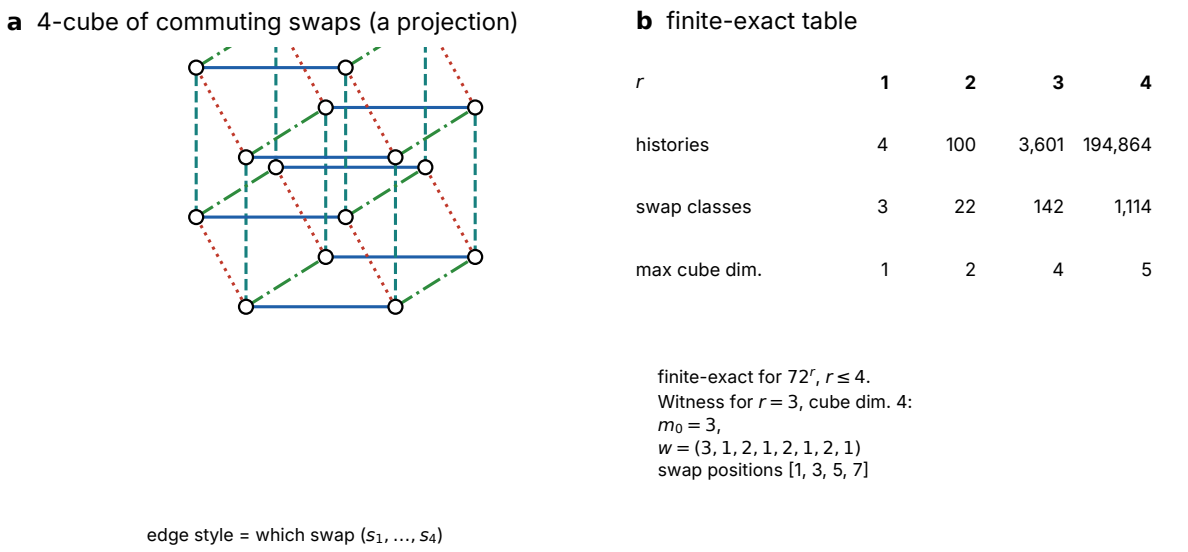


Figure 14: **Commuting swaps and cubical faces (finite-exact data; the cube is drawn in projection).** (a) A projection of the 4-cube formed by the sixteen histories obtained from the witness of Proposition 8.3 by performing any subset of its four commuting exchanges; the four edge styles are the four exchanges $s_1, \ldots, s_4$. It is a combinatorial 4-cell and is not embedded in a Euclidean space of dimension 4 by the drawing. (b) The table of table 2.

8.2 The histories of 72^r: classes, groups and homology

The figure above is Figure 14; its caption states the exactness of what is drawn. For $r = 3$ the source-and-content groups number 141 and the swap classes 142: exactly one group of histories (source 15, content $\{2, 4, 5, 5, 15\}$, endpoint 72^3, two classes of four histories each) splits into two classes; for $r \leq 2$ classes and groups coincide. Over all non-squarefree endpoints $t \in [4, 6000]$ with $H(t) \leq 4000$ (2 354 endpoints), the endpoints at which some group of histories splits are exactly $1944, 2400, 4050, 4320, 5400$ (the first two sources of the pattern: source 2, content $\{1, 2, 3, 8\}$, endpoint 1944, two classes of sizes 4 and 1; and the family of Example 5.7, endpoint 5400). FINITE-EXACT.

Source record: `D-T6` (`CW-0343`); status FINITE-EXACT; treatment FINITE.

Remark 8.5 (cube dimension is not homological degree). The cube of Theorem 8.2 is contractible, and a d-cube of commuting swaps contributes no homology of degree d by its presence. For $r = 3$ the complex Q has cubes of dimension up to 4 (four of them) but $\mathbb{F}_2$-Betti numbers $(142, 368, 3, 0, 0)$: homology lives in degrees $0, 1, 2$ only. The combinatorial cube dimension is an upper bound on the dimension of a cell, not an indication of any homological degree.

Proposition 8.6 (finite-exact: the groups are not always the classes)**.** *The histories* $(15; 5, 4, 5, 2, 15)$ *and* $(15; 15, 2, 4, 5, 5)$ *have the same source, content* $\{2, 4, 5, 5, 15\}$ *and endpoint* 72^3 *and lie in different swap classes; likewise* $(2; 1, 2, 3, 8)$ *and* $(2; 1, 8, 3, 2)$ *(endpoint* 1944*).*

8.3 Dimension of an obstruction

Let C be a set of distinct labels and m a source. Three numbers are attached to (C, m): $d_{\text{span}} = \text{rank}_{\mathbb{Q}}\{\Delta(N) : N \in C\}$ (the rank of the displacement vectors over *all* primes); $d_{\text{con}} =$ the number of primes p such that relaxing the constraint at p (replacing $v_p(m)$ by ∞, i.e. dropping p from the thresholds) changes $\mathcal{A}(m)$; and $d_{\text{top}} =$ the least size of a set S of primes such that keeping only the constraints at S preserves the Betti vector of K_m.

Theorem 8.7 (passive coordinates)**.** *A prime that divides no label of* C *never constrains:* $\Delta_p(N) = 2v_p(N+1) \geq 0$ *for* $N \in C$*, so* $b_p(w) = 0$ *for every word* w *over* C*. Hence admissibility, and every complex built from it, depends only on the active primes* $\{p : p \mid N \text{ for some } N \in C\}$*.*

Proof. Immediate from Theorem 4.2: $S_j(p) \geq 0$ for all j. □

Source record: `B-T1` (`CW-0383`); status PROVED; treatment FULL.

Proposition 8.8 (the three dimensions differ)**.** *For* $\{2, 3, 8, 9\}$ *at* 72*:* $(d_{\text{span}}, d_{\text{con}}, d_{\text{top}}) = (3, 2, 2)$*; for* $\{2, 3, 8, 9, 14\}$ *at* 1008*:* $(4, 2, 2)$*. In the first case the displacement vectors live in the primes* $2, 3, 5$ *(since* $9 + 1 = 10$*) and the active primes are* $2, 3$*; in the second the displacement span has rank* 4 *in the primes* $2, 3, 5, 7$*, the active primes are* $2, 3, 7$*, the prime* 7 *never constrains at* 1008 *(*$v_7(1008) = 1$ *and only the label* 14 *consumes it), and the obstruction (the void of Example 7.9) is already carried by the constraints at* $\{2, 3\}$*.*

Proof. Explicit exact computation: the ranks of the Δ-matrices are 3 and 4; relaxing 2 or 3 changes $\mathcal{A}(m)$ and relaxing 7 does not; keeping only the constraints at $\{2, 3\}$ reproduces the Betti vectors $(1, 1, 0)$ and $(1, 0, 1, 0)$, and no single prime does. See Appendix B. □

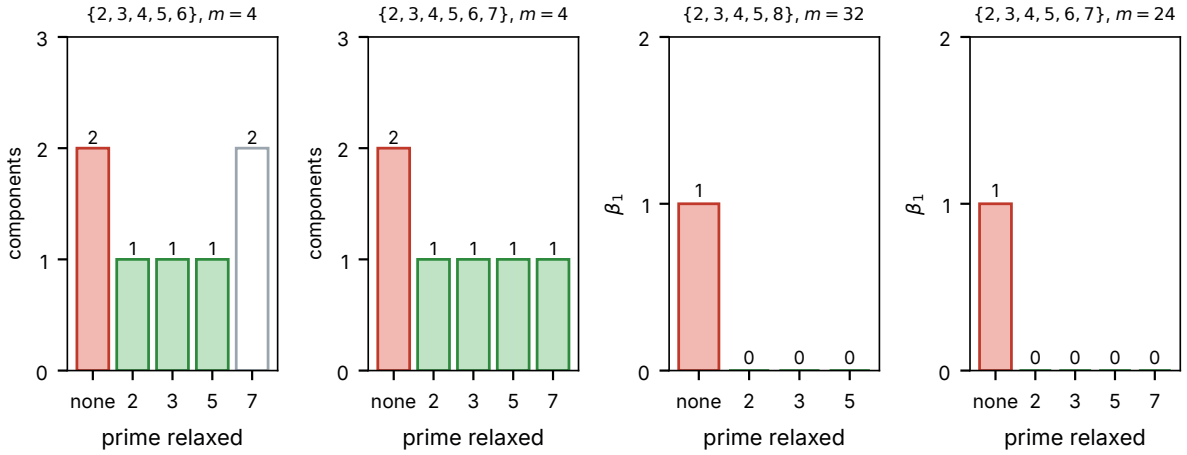


Figure 15: **Irreducible prime-resource obstructions (exact).** For each of the four instances of Proposition 8.9: the number of components (panels 1–2) or the first Betti number (panels 3–4) of K_m when nothing is relaxed (red bar) and when exactly one prime is relaxed (green bars). In panel 1 relaxing the passive prime 7 (which does not constrain, by Theorem 8.7) is also shown and leaves 2 components. Every active prime, relaxed alone, removes the obstruction. Theorem illustrated: Proposition 8.9, Theorem 8.7.

Source record: `B-T2` (`CW-0384`); status PROVED; treatment FULL.

A rank-4 valuation realisation therefore carries a void whose obstruction is 2-dimensional in the sense of constraints: the rank of the valuation span is not an obstruction dimension.

Proposition 8.9 (irreducible multi-prime obstructions)**.** *Each of the following four statements about the route complex K_m is an exact finite fact (replayed in* `checks/`*, with the admissible sets listed in Appendix B). For the content C and source m below, K_m has the stated feature; and relaxing the constraint at any* single *active prime makes K_m contractible (Betti vector $(1, 0, \ldots, 0)$), so every active prime is essential and the obstruction is genuinely* 3- *(resp.* 4-*)dimensional.*

feature	*content C*	*source m*	*active primes (all essential)*
2 components	$\{2, 3, 4, 5, 6\}$	4	$2, 3, 5$
2 components	$\{2, 3, 4, 5, 6, 7\}$	4	$2, 3, 5, 7$
loop ($\beta_1 = 1$)	$\{2, 3, 4, 5, 8\}$	32	$2, 3, 5$
loop ($\beta_1 = 1$)	$\{2, 3, 4, 5, 6, 7\}$	24	$2, 3, 5, 7$

Source record: `B-T3` (`CW-0385`); status PROVED; treatment FULL.

Irreducible means: relaxing any one active prime removes the obstruction. In these examples the obstruction is a conjunction of independent single-prime blockers (every single relaxation trivialises it; none is a synergy of several primes). The number of active primes is not the dimension of the homology: the loop obstructions have first Betti number 1 whatever their prime count, and the void has degree 2 with constraint dimension 2. The figure above is Figure 15; its caption states the exactness of what is drawn.

Proposition 8.10 (census of obstruction support; finite-exact, cited)**.** *In the programme's census of route complexes with $k \le 5$ labels (cap of 500 resource vectors per content), disconnections occur with support $2, 3, 4$; loops with support $2, 3$ (never 4 for $k \le 5$); voids only with support 2; and the cut order (the least number of simultaneously relaxed primes that makes the complex contractible) is 1 in about 17 900 non-trivial feature instances. This was not re-run here. Whether synergistic obstructions of cut order ≥ 2 exist is open.*

Source record: `B-T4` (`CW-0386`); status FINITE-EXACT; treatment FINITE.

Problem 8.11 (open: repeated labels)**.** The complexes of this section and of Section 7 are defined for sets of distinct labels. For repeated labels the face structure of the arrangement graph needs a construction by chains of sub-multisets; none is given here, and Theorems 7.3 to 7.5 are not claimed for repeated labels. (The prefix-set form, the threshold law and Theorem 6.3 do hold for repeated labels.)

9 Different-content endpoint relations

Up to now the content of a history, the multiset of its labels, was fixed. Here two histories with *different* contents are compared. They can meet at one endpoint only through an arithmetic relation among the multipliers $R(N)$, and this relation is a lattice kernel.

9.1 The valuation matrix and its kernel

Fix a finite alphabet A of labels. Let $\mathcal{P}$ be the set of primes dividing some N or $N+1$ with $N \in A$, and let $V_A = (\Delta_p(N))_{p \in \mathcal{P},\, N \in A}$ be the integer matrix of displacement vectors. For a history h over A let $c(h) \in \mathbb{N}^A$ be its content vector.

Theorem 9.1 (endpoint equality is a kernel condition)**.** *Two histories over A from the same source have the same endpoint if and only if $V_A(c - c') = 0$.*

Proof. The endpoint is $m \prod_N R(N)^{c_N}$. A positive rational is determined by its p-adic valuations, $v_p(R(N)) = \Delta_p(N)$, and a prime outside $\mathcal{P}$ has valuation 0 for every $R(N)$, $N \in A$. Hence the endpoints agree iff $\sum_N (c_N - c'_N)\Delta_p(N) = 0$ for all $p \in \mathcal{P}$. □

Source record: `D4-T1` (`CW-0369`); status PROVED; treatment FULL.

Finite check. Over all words of length ≤ 6 from four sources, and on a box of 8 110 682 content pairs of the programme's records, there is no violation (cited; the statement is a theorem, the check only guards the implementation). The content vectors of equal-endpoint histories form a coset of $\ker_{\mathbb{Z}} V_A$ intersected with $\mathbb{N}^A$ (a *toric fibre*).

Theorem 9.2 (every kernel vector is realised)**.** *Let $0 \ne z \in \ker_{\mathbb{Z}} V_A$, $z = z^+ - z^-$ with $z^{\pm} \ge 0$, and let $m = \prod_N N^{|z_N|}$. Then every arrangement of the content z^+ and every arrangement of the content z^- is admissible from m, and all of these histories have one endpoint, namely $m \prod_N R(N)^{z^+_N} = m \prod_N R(N)^{z^-_N}$. For arrangements u of z^+ and v of z^- the pair is simultaneously admissible from a source m' iff $\operatorname{lcm}(K(u), K(v)) \mid m'$; the minimal sources of the relation form the antichain of the minimal $\operatorname{lcm}(K(u), K(v))$.*

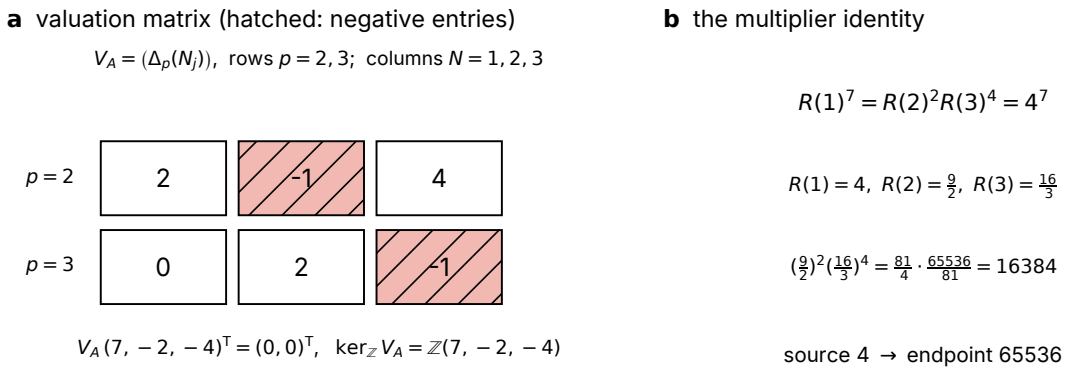


Figure 16: **The valuation matrix of $\{1, 2, 3\}$ and the multiplier identity (exact).** (a) The matrix $V_A = (\Delta_p(N))$ with rows $p = 2, 3$ and columns $N = 1, 2, 3$; entries are exact integers, negative entries are shaded and hatched. $V_A(7, -2, -4)^{\mathsf{T}} = 0$ and the kernel over $\mathbb{Z}$ is $\mathbb{Z}(7, -2, -4)$. (b) The identity $R(1)^7 = R(2)^2 R(3)^4 = 4^7$ behind it; source 4, endpoint 65536. Theorem illustrated: Theorem 9.3.

Proof. The content z^+ has $P(z^+) = \prod N^{z_N^+}$ dividing m, and likewise $P(z^-)$; by Proposition 4.6 all arrangements are admissible. The endpoints coincide by Theorem 9.1. The simultaneous admissibility criterion is Theorem 4.2 applied to u and v. □

Source record: `D4-T2` (`CW-0370`); status PROVED; treatment FULL.

The realising source is sufficient, not minimal. For the kernel vector z below the pair $(2, 2, 3, 3, 3, 3)$ against (1^7) needs only the source 4, whereas $\prod N^{|z_N|} = 1 \cdot 2^2 \cdot 3^4 = 324$.

Theorem 9.3 (the labels $\{1, 2, 3\}$)**.** $\ker_{\mathbb{Z}} V_{\{1,2,3\}} = \mathbb{Z}(7, -2, -4)$*; equivalently* $R(1)^7 = R(2)^2 R(3)^4 = 4^7 = 16384$.

Proof. Only the primes $2, 3$ occur: $R(1) = 4$, $R(2) = 9/2$, $R(3) = 16/3$ give

$$V = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 2 & -1 \end{pmatrix} \quad (\text{rows } p = 2, 3;\ \text{columns } N = 1, 2, 3).$$

The rows are independent, so the kernel has rank 1. From the second row $c_3 = 2c_2$ and from the first $2c_1 - c_2 + 8c_2 = 0$, i.e. $c_1 = -\frac{7}{2}c_2$; the primitive generator is $(7, -2, -4)$. Direct evaluation: $(9/2)^2(16/3)^4 = \frac{81}{4} \cdot \frac{65536}{81} = 16384 = 4^7$. □

Source record: `D4-T4` (`CW-0372`); status PROVED; treatment FULL.

Example 9.4 (the benchmark merger from 4)**.** From the source 4 the history 1^7 passes through $4, 16, 64, 256, 1024, 4096, 16384, 65536$. The content $\{2, 2, 3, 3, 3, 3\}$ has exactly three admissible arrangements from 4, namely $(2, 2, 3, 3, 3, 3)$ through $4, 18, 81, 432, 2304, 12288, 65536$, and the arrangements $(2, 3, 2, 3, 3, 3)$, $(2, 3, 3, 2, 3, 3)$ (states shown in Figure 17). All end at $4^8 = 65536$. Here $K(1^7) = 1$, $K(2, 2, 3, 3, 3, 3) = 4$ and the smallest threshold of the three arrangements is 2. FINITE-EXACT in the arrangement count.

The figure above is Figure 16; its caption states the exactness of what is drawn.

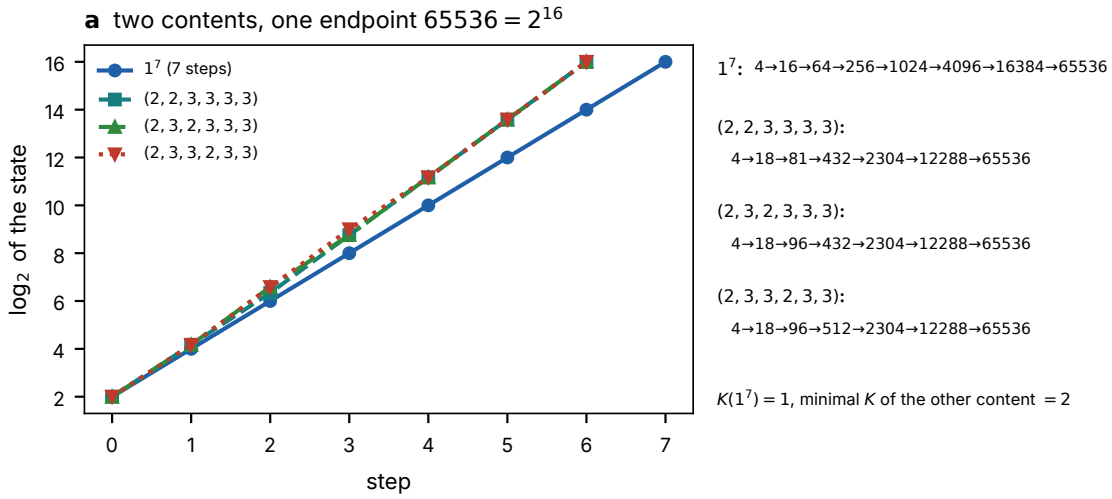


Figure 17: **Two contents with one endpoint (exact, source 4, endpoint $65536 = 2^{16}$).** (a) $\log_2$ of the state against the step for the history 1^7 (circles, 7 steps) and the three admissible arrangements of $\{2^2, 3^4\}$ (squares, triangles; 6 steps; dashed, dash-dotted and dotted lines respectively). The curves are different but meet at the endpoint. (b) The exact state sequences. $K(1^7) = 1$; the smallest threshold among the three arrangements is 2. Theorem illustrated: Theorem 9.2, Theorem 9.3.

9.2 Geometry of the recombination: traces, area and closure

Write $d_N = \Delta(N) \in \mathbb{Z}^{\mathcal{P}}$ and let the *trace* of a history be the set of its visited states.

Theorem 9.5 (trace and area)**.** *(a)* Trace injectivity. *Distinct histories from one source have distinct traces: the states strictly increase ($R(N) > 1$), so ordering the trace recovers the state sequence, and consecutive ratios recover the labels because R is injective.*

(b) Area potential. *For a word $w = (N_1, \dots, N_l)$ put $\Omega(w) = \sum_{i<j} d_{N_i} \wedge d_{N_j}$ (a bivector). An adjacent exchange of $N_k = a$ and $N_{k+1} = b$ changes Ω by exactly $-2\, d_a \wedge d_b$ (sharpening the factor 1 recorded in the source); a block replacement $u \to v$ with $\sum d_u = \sum d_v$ changes Ω by $\Omega(v) - \Omega(u)$, independently of the prefix and the suffix. For arrangements of $\{2^2, 3^4\}$ with* inv *the number of pairs with a 3 before a 2,*

$$\Omega = \det(d_2, d_3)\,(8 - 2\,\mathrm{inv}) \ \text{(in the } (v_2, v_3)\text{-plane)}, \quad \det(d_2, d_3) = -7.$$

(c) Closure triangle. *From the source 4 the two histories 1^7 and $(2, 2, 3^4)$, in the (v_2, v_3)-plane, bound the triangle with vertices $(2, 0), (0, 4), (16, 0)$ (sides $2d_2$, $4d_3$ and $7d_1$) of area $28 = 4\,|\det(d_2, d_3)|$; the two paths meet only at their endpoints.*

Proof. (a) was argued above. (b) For an exchange, the terms $d_{N_i} \wedge d_{N_k} + d_{N_i} \wedge d_{N_{k+1}}$ with $i < k$ and $d_{N_k} \wedge d_{N_j} + d_{N_{k+1}} \wedge d_{N_j}$ with $j > k+1$ are symmetric in the exchanged pair, and the single pair term $d_a \wedge d_b$ becomes $d_b \wedge d_a = -d_a \wedge d_b$, which changes Ω by $-2d_a \wedge d_b$. For a block replacement, the cross terms between a prefix or suffix and the block are of the form $d_N \wedge \sum d_u$, and $\sum d_u = \sum d_v$. For $2^2 3^4$: pairs of equal labels contribute 0; a pair 2 before 3 contributes $d_2 \wedge d_3 = -7$ and a pair 3 before 2 contributes $+7$, so $\Omega = -7(8 - \mathrm{inv}) + 7\,\mathrm{inv}$. (c) The trace of 1^7

is $(2,0),(4,0),\ldots,(16,0)$; that of $(2,2,3,3,3,3)$ is $(2,0),(1,2),(0,4),(4,3),(8,2),(12,1),(16,0)$ (exact, verified from the state values). The triangle's area is $\frac{1}{2}\cdot 14\cdot 4 = 28$; the second path stays in $v_3 > 0$ except at its endpoints. □

Source record: `B-T5` (`CW-0387`); status PROVED; treatment FULL.

Remark 9.6 (what Ω is and is not). Ω is a function on vertices of the swap-and-rewrite graph, so the signed area of every closed combinatorial loop in that graph is 0 by construction; it measures the *area between two histories*, a weighted inversion statistic, not a homology class. It realises the 1-skeleton and the bigons of the extended complex; higher cells are not realised as solids, and no coherence cells are present.

Proposition 9.7 (swap diamonds are nondegenerate; finite-exact)**.** *No two distinct labels $N < M \le 400$ have parallel displacement vectors $d_N \parallel d_M$, so every swap diamond has nonzero area in some coordinate plane. The statement for all N is open.* FINITE-EXACT *(recomputed here: 0 parallel pairs among labels $1,\ldots,400$).*

Source record: `B-T6` (`CW-0388`); status FINITE-EXACT; treatment FINITE.

9.3 The cube family

Let $\mathrm{A} = 1^7$ and $\mathrm{B} = 2^2 3^4$ be the two sides of the relation of Theorem 9.3. Starting at 4, take r successive blocks, each independently A or B.

Theorem 9.8 (block cube family)**.** *The 2^r histories so obtained all end at 4^{7r+1}, have the $r+1$ contents $\mathrm{A}^{r-j}\mathrm{B}^j$ $(0 \le j \le r)$, and the cubical complex on the 2^r histories whose cells are the sets of independent block exchanges has $C(r,d)2^{r-d}$ cells of dimension d and is contractible (the r-cube).*

Proof. Each block multiplies by 4^7 (via A) or $R(2)^2R(3)^4 = 4^7$ (via B) by Theorem 9.3; the endpoint is $4\cdot 4^{7r}$. A choice of blocks is a word in $\{\mathrm{A},\mathrm{B}\}^r$; the cell count is that of the r-cube, which is contractible. The state before every block is a power of 4, hence a multiple of $K(2,2,3,3,3,3) = 4$, so the block B is admissible there by Theorem 4.2; the block $\mathrm{A} = 1^7$ is admissible from every state. □

Source record: `D4-T5` (`CW-0373`); status PROVED; treatment FULL.

(The cell counts coincide with those of the cube; the homology is classical. The programme's verifier records Betti vectors $(1,0,\ldots,0)$ for $r \le 4$ FINITE-EXACT.)

9.4 Endpoint information and the kernel of content ambiguity

For a fixed source m and alphabet A, a history of content c ends at t with $v(t) - v(m) = V_A c$, where v denotes the vector of valuations at the primes $\mathcal{P}$ (Theorem 9.1). The endpoint therefore determines the *image* $V_A c$, and two content vectors give the same endpoint exactly when they differ by an element of $\ker_{\mathbb{Z}} V_A$: the kernel coset is the content ambiguity of the endpoint. The rank of V_A is a property of the alphabet and not of an individual endpoint. For the alphabet $\{1,2,3,8\}$ the matrix V has rank 2 and its kernel has rank 2. The *Graver candidates* found in the box of side 8 are

$$z_1 = (1,-4,0,2),\ z_2 = (7,-2,-4,0),\ z_3 = (3,1,-2,-1),$$
$$z_4 = (4,-3,-2,1),\ z_5 = (2,5,-2,-3)$$

FINITE-EXACT as a Graver list; the box is not a proof of completeness of the Graver basis (Proposition 10.7). What the endpoint does not determine, among contents, is the position within the kernel coset; block rewrites are the moves through that coset (Section 10).

10 Block rewrites, source exposure and resource constraints

10.1 The rewrite grammar and its threshold

Definition 10.1 (rewrite)**.** Let $z \in \ker_{\mathbb{Z}} V_A$ with $z = z^+ - z^-$, and let u and v be arrangements of the contents z^+ and z^-. A *block rewrite* is the replacement $p \cdot u \cdot s \to p \cdot v \cdot s$ of a contiguous block, where p, s are words. Because $R(u) = R(v)$ the state after $p \cdot u$ equals the state after $p \cdot v$, so the suffix s and the endpoint are unchanged. The *rewrite graph* of a source m, an endpoint and a move set M of kernel vectors has the admissible histories as vertices and the legal swaps and the legal block rewrites with $z \in \pm M$ as edges; its components are the *rewrite classes*. A rewrite class is a union of swap classes.

Theorem 10.2 (block threshold)**.** *Let $p \cdot u \cdot s$ be admissible from m. Then $p \cdot v \cdot s$ is admissible from m iff v is admissible from the state x_p reached after p, i.e. iff* $\operatorname{lcm}(K(pus), K(pvs)) \mid m$. *The rewrite is thus an event local to the prefix state x_p: its executability depends on x_p only through the valuations the block needs.*

Proof. The prefix p is common and admissible. After $p \cdot v$ the state is the same as after $p \cdot u$, so the suffix s is admissible from it; the only remaining condition is the admissibility of v from x_p. The equivalence with the lcm is the threshold law applied to the two words. □

Source record: `D4-T3` (`CW-0371`); status PROVED; treatment FULL.

Check. 425 random cases of the criterion were replayed in this work, with 0 violations (cited: 11 000 randomized tests in the programme's records).

10.2 Saturation and the quotient description

Theorem 10.3 (saturation)**.** *Suppose that the contents of the histories from m to a fixed endpoint are connected by moves of a set M (a Graver basis suffices) staying inside the fibre, and that $P(c) \mid m$ for every content c. Then the histories form a single rewrite class.*

Proof. By Proposition 4.6 every arrangement of every content is admissible, so each content is one swap class. For a move z taking c to $c' = c + z$, take any history of c, swap it into the form $u \cdot r$ with u an arrangement of z^- and r the rest; then $v \cdot r$ is admissible by the same proposition and joins the two swap classes. Connectivity of the fibre under M now gives connectivity of the rewrite graph. (A Graver basis is a Markov basis, [12, 13].) □

Source record: `D4-T6a` (`CW-0374`); status PROVED; treatment FULL.

Proposition 10.4 (quotient description)**.** *The rewrite classes are exactly the connected components of the graph whose nodes are the swap classes and whose edges are the* exposed redexes*: a redex u or v occurring as a contiguous block in some word of a class whose replacement is admissible.*

Proof. By Definition 10.1: a rewrite edge between two histories is by definition an exposed redex, and the classes of the graph with swap edges and rewrite edges are the components of the quotient by swap classes. □

Source record: `D4-T6b` (`CW-0375`); status PROVED; treatment FULL.

Corollary 10.5 (three obstruction types)**.** *Below the saturation condition a rewrite graph can be disconnected for three separately observable reasons: (a)* algebraic*, the content fibre is disconnected under M; (b)* exposure*, contents are linked by a move of M but no history has a contiguous redex because the context labels cannot be moved out of the block; (c)* emptiness*, some content has no admissible history from m.*

Algebraic obstructions hold at every source and are the subject of Proposition 10.7; exposure and emptiness are source-dependent and are separated for a test family below.

10.3 Positive and negative completeness

Proposition 10.6 (three labels; finite-exact)**.** *For the alphabet $\{1,2,3\}$, swaps together with the single move $1^7 \leftrightarrow 2^2 3^4$ connect all histories of equal source and endpoint, in the box $v_2(m) \le 1$, $v_3(m) \le 4$, $c_3 \le 8$ with fibre length ≥ 2 (30 fibres, 840 fibre-source pairs, no disconnected pair; 16 fibres and 364 pairs re-checked independently in the programme). General m is open: the fibres are the rank-one chains $c + t(7,-2,-4)$, so a proof would need an exposed redex for every consecutive pair of contents.* FINITE-EXACT.

Source record: `D4-T6c` (`CW-0376`); status FINITE-EXACT; treatment FINITE.

Proposition 10.7 (the Graver basis cannot be omitted)**.** *On the alphabet $\{1,2,3,8\}$ let $z_1 = (1,-4,0,2)$ and $z_2 = (7,-2,-4,0)$. The fibre $\{3^2 8,\ 1^3 2\}$ at endpoint $288 = R(3)^2 R(8) = R(1)^3 R(2)$ contains two contents that are* not *joined by $\pm z_1, \pm z_2$ and swaps: neither content admits a move (the contents $(0,0,2,1)$ and $(3,1,0,0)$ have $c_2 \le 1$, $c_3 \le 2$, $c_1 \le 3$, so none of $c \pm z_1$, $c \pm z_2$ is nonnegative). They are joined by the Graver vector $z_3 = (3,1,-2,-1) = \big((3,1,0,0) - (0,0,2,1)\big)$.* PROVED *for the explicit pair; the programme's census finds 498 of 816 fibres deficient for $\{z_1, z_2\}$* FINITE-EXACT*, not re-run here.*

Source record: `D4-T7c` (`CW-0379`); status PROVED; treatment FULL.

So generating the kernel is not enough: the move set must contain a Graver (Markov) basis for the fibre to be algebraically connected.

10.4 An exposure dichotomy for all j

For $j \ge 1$ consider the family $U_j = \{2^4, 3^j\}$, $V_j = \{1, 8^2, 3^j\}$ with the move $2222 \leftrightarrow$ any arrangement of $\{1,8,8\}$ (the kernel vector z_1); $R(2)^4 = R(1)R(8)^2 = 3^8/2^4$. Write $a = v_2(m)$, $b = v_3(m)$.

Theorem 10.8 (exposure dichotomy)**.** *For every $j \ge 1$: both contents have admissible histories from m iff $b + 8 \ge j$ and $(a,b) \ne (0,0)$. In that case the swap-and-rewrite space is connected iff $b \ge 1$ or $a \ge 4$ (i.e. iff $3 \mid m$ or $16 \mid m$), and otherwise it has exactly two components, one swap class per content.*

Proof. Realisability. Each 2 adds 2 to v_3, each 8 adds 4, and each 3 removes 1, so the final v_3 is $b+8-j \geq 0$. If $a = b = 0$, then U_j cannot start (2 needs $v_2 \geq 1$ and 3 needs $v_3 \geq 1$) and V_j stalls after the leading 1 (8 needs $v_2 \geq 3$, 3 needs $v_3 \geq 1$; a leading 1 gives $v_2 = 2$ only). Otherwise explicit words exist: for $b \geq 1$, $3, 2, 2, 2, 2, 3^{j-1}$ and $3, 1, 8, 8, 3^{j-1}$; for $b = 0, a \geq 1$ (so $j \leq 8$), $2, 3, 2, 2, 2, 3^{j-1}$ and $1, 8, 3, 8, 3^{j-1}$.

One swap class per content. Let A, B be coprime labels such that $2v_p(A+1) \geq v_p(B)$ for $p \mid B$ and $2v_q(B+1) \geq v_q(A)$ for $q \mid A$ (here $(2,3)$ and $(8,3)$: $2v_3(3) = 2 \geq 1$, $2v_2(4) = 4 \geq 1$, $2v_3(9) = 4 \geq 1$). Claim: from any state, all admissible orderings of a multiset $\{A^s, B^t\}$ lie in one swap class. Induct on the length. Take admissible $A \cdot r$ and $B \cdot r'$. In $A \cdot r$ the first B occurs after some A's; it can be moved to the front of r by legal adjacent swaps: at the state after the first A we have $v_p \geq 2v_p(A+1) \geq v_p(B)$ for $p \mid B$, so B is legal there; the A's it passes only gain resources, since $\Delta_q(B) = 2v_q(B+1) \geq 0$ for $q \mid A$ and $q \nmid B$. Thus $A \cdot r$ is in the class of $A \cdot B \cdot r''$. Symmetrically $B \cdot r'$ is in the class of $B \cdot A \cdot r'''$. Both orders AB and BA are legal from the start state (each letter is legal at the start by assumption, and the other raises the resources of the first), with the same state afterwards; by induction the arrangements after them lie in one class, so the two classes coincide. For V_j the label 1 can be moved to the front by legal swaps (it divides everything and only raises v_2), so the claim applies after it.

Connectivity. If $b \geq 1$, both $3 \cdot 2222 \cdot 3^{j-1}$ and $3 \cdot (1,8,8) \cdot 3^{j-1}$ are admissible: after the leading 3 the valuation $v_2 \geq 4$, the blocks leave $v_3 = b + 7 \geq j - 1$, and for $(1,8,8)$ the walk of v_2 is $\geq 6, \geq 3, \geq 0$. They have one endpoint, so the block rewrite joins the two swap classes. If $a \geq 4$, $b = 0$ (so $j \leq 8$), then $2222 \cdot 3^j$ and $1 \cdot 8 \cdot 8 \cdot 3^j$ are admissible and the same argument applies.

Obstruction ($3 \nmid m$, $a \leq 3$, $(a,b) \neq (0,0)$). A label 3 is applicable only after v_3 has been raised by a 2 or an 8. In U_j all 2s lie in the redex 2222, in V_j all 8s lie in the redex (an arrangement of $\{1,8,8\}$). So no 3 precedes a redex, the redex starts at position 0 with $v_2 = a \leq 3$, and: 2222 needs $v_2 \geq 4$; the arrangements $(1,8,8)$, $(8,1,8)$ need $a \geq 4$, $(8,8,1)$ needs $a \geq 6$. No redex is ever exposed, so the two swap classes are not joined. □

Source record: `B-T9` (`CW-0391`); status PROVED; treatment FULL.
The figure above is Figure 18; its caption states the exactness of what is drawn.

Remark 10.9 (supersession and sharpness). The source release D4-T7b stated the dichotomy for $j \leq 8$ (finite-exact) and marked general j open; the addendum proves it for all j (the sufficient direction above, and the realisability bound $j \leq b + 8$, are what made the finite range natural). The saturation bound of Theorem 10.3 for this family is $2^6 3^j$ ($P(U_j) = 2^4 3^j$, $P(V_j) = 2^6 3^j$) while the true connectivity boundary is $3 \mid m$ or $16 \mid m$: the product bound is not sharp, and the sharp boundary is an exposure condition, not an all-arrangements condition.

Source record: `D4-T7a` (`CW-0377`); status PROVED; treatment FULL.

Source record: `D4-T7b` (`CW-0378`); status MIXED; treatment SPLIT.

We recomputed the dichotomy for $j \leq 6$ over all (a,b) with $a \leq 6$, $b \leq j$ in this work, and it agrees (183 rows, no violation).

10.5 Content-changing rewrites can create loops

Example 10.10 (the join example; exact)**.** $R(2)R(8)R(63) = R(5)R(9)R(35) = 20736/7$. At the sources $m = 280$ and $m = 840$ the four histories $(2, 63, 8)$, $(8, 63, 2)$, $(5, 9, 35)$, $(35, 9, 5)$ have pairwise different traces; there is no legal swap, so the swap space is 4 points with 4 swap classes.

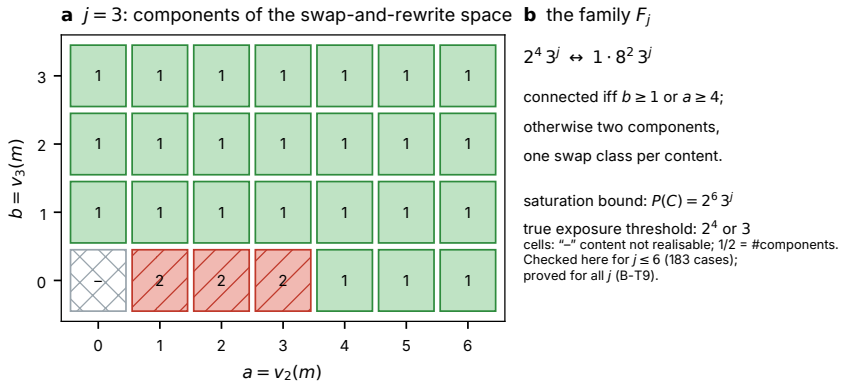


Figure 18: **Components of the swap-and-rewrite space of $2^4 3^j \leftrightarrow 1 \cdot 8^2 3^j$ for $j = 3$ (finite-exact table, matching Theorem 10.8).** The entry at $(a, b) = (v_2(m), v_3(m))$ is the number of components: 0 (nothing realisable) at $(0, 0)$, 2 at $b = 0$ and $a \in \{1, 2, 3\}$ (exposure obstruction), 1 elsewhere. Colour is redundant with the printed integers. Theorem illustrated: Theorem 10.8.

The block replacements $(2, 63, 8) \leftrightarrow (5, 9, 35)$, $(2, 63, 8) \leftrightarrow (35, 9, 5)$, $(8, 63, 2) \leftrightarrow (5, 9, 35)$, $(8, 63, 2) \leftrightarrow (35, 9, 5)$ are all admissible and form a 4-cycle: one rewrite class, cells $\{0 : 4, 1 : 4\}$, $\mathbb{F}_2$-Betti vector $(1, 1)$. A content-changing relation creates a loop out of a disconnected swap space.

Source record: `D4-T8a` (`CW-0380`); status PROVED; treatment FULL.

Problem 10.11 (open: homotopy type of extended complexes)**.** The extended complex of histories with swaps and block rewrites has no coherence cells: braid hexagons and relations between overlapping moves are not filled, so its Betti numbers are invariants of this specific complex only. Its homotopy type, the effect of adding coherence cells, and Künneth-type behaviour for products of loop sides are open. A cycle in a restricted complex need not be a cycle in the complete one. OPEN.

Source record: `D4-T8b` (`CW-0381`); status OPEN; treatment OPEN.

11 Occurrence-event invariants and prime-sensitive obstructions

11.1 The occurrence fibre

For an integer state x let $D(x) = \{d \geq 2 : d^2 \mid x\}$ be its *occurrence fibre*. A history $h = (x_0, x_1, \ldots, x_l)$ induces the *fibre sequence* $D(x_0), \ldots, D(x_l)$, and for each index $d \geq 2$ the incidence sequence $\rho_d(h) = ([d \in D(x_i)])_i \in \{0, 1\}^{l+1}$.

Lemma 11.1 (orthant form)**.** *$d \in D(x)$ iff $v_p(x) \geq 2v_p(d)$ for every prime $p \mid d$. Hence $D(x) = D(x')$ iff $\lfloor v_p(x)/2 \rfloor = \lfloor v_p(x')/2 \rfloor$ for all p, and ρ_d is the indicator of the valuation walk lying in the shifted orthant $O_d = \{v : v_p \geq 2v_p(d)\ \forall p \mid d\}$; for a prime power p^k it is the half-space $v_p \geq 2k$.*

Source record: `OCCURRENCE_BRIDGE:Orthant-form` (`CW-0356`); status PROVED; treatment FULL.

Lemma 11.2 (labels are visible)**.** *If $A \neq B$ are both admissible at x then $D(T_A x) \neq D(T_B x)$. More precisely some prime p has $|v_p(T_A x) - v_p(T_B x)| \geq 2$, so p^k lies in exactly one of the two fibres for suitable k.*

Proof. Since $A + 1 \neq B + 1$ some prime p has, say, $v_p(A + 1) > v_p(B + 1) \geq 0$. Then $p \nmid A$ and $\Delta_p(A) = 2v_p(A + 1) \geq 2$. If $v_p(B + 1) \geq 1$ then $\Delta_p(B) = 2v_p(B + 1)$ and the difference is ≥ 2. If $v_p(B + 1) = 0$ then $\Delta_p(B) = -v_p(B) \leq 0$ and the difference is $\geq 2v_p(A + 1) \geq 2$. A difference ≥ 2 in v_p gives different integer parts of $v_p/2$. □

Theorem 11.3 (faithfulness, locality, event locality)**.** *(a) For a fixed source the fibre sequence $F(h)$ determines the history h. (b) A legal swap changes exactly one entry of $F(h)$; hence the occurrence signature is never a swap invariant, and swaps preserve exactly the source, content, endpoint and all but one fibre. (c) If d is lost at a step with label N then $\gcd(d, N) > 1$; if d is born then $\gcd(d, N + 1) > 1$; the label 1 never causes a loss.*

Proof. (a) Induction on length: given x_{j-1}, Lemma 11.2 says the fibre $D(x_j)$ determines N_j among the labels admissible at x_{j-1}. (b) A swap at position i changes only the intermediate state; the two intermediate states are $T_A x$ and $T_B x$ with different fibres by Lemma 11.2. (c) If $d \in D(x) \setminus D(Tx)$ there is $p \mid d$ with v_p falling below $2v_p(d)$, so $\Delta_p(N) < 0$, i.e. $p \mid N$. If d is born then $\Delta_p(N) > 0$ for some $p \mid d$, so $p \mid N + 1$. Finally $\Delta_p(1) = 2v_p(2) \geq 0$. □

Source record: `D-T4` (`CW-0341`); status PROVED; treatment FULL.

Corollary. Histories with the same source and endpoint have pairwise different fibre sequences even when they are swap-equivalent: the endpoint keeps only the last entry. In the benchmark of 72, $\rho_2(h_1) = (0, 1, 1, 1)$ and $\rho_2(h_2) = (0, 1, 0, 1)$ for the two swap-equivalent histories. Finite-exact check: 0 collisions among 104 histories and among 13 950 label pairs.

Proposition 11.4 (local discrepancy)**.** *For a legal swap with intermediate states $a = T_A x$, $b = T_B x$ and half-valuation vectors $e_p = \lfloor v_p/2 \rfloor$, the changed entry differs by*

$$|D(a) \triangle D(b)| = \prod_p (e_p(a) + 1) + \prod_p (e_p(b) + 1) - 2 \prod_p (\min(e_p(a), e_p(b)) + 1) > 0,$$

because $D(a) \cap D(b) = D(\min(e(a), e(b)))$ and $|D(x)| = \prod_p (e_p(x) + 1) - 1$. The swap preserves ρ_d iff $d \notin D(a) \triangle D(b)$, i.e. iff a and b lie on the same side of the orthant O_d.

Source record: `OCCURRENCE_BRIDGE:Local-discrepancy` (`CW-0357`); status PROVED; treatment FULL.

11.2 Persistence-restricted swap connectivity

For an index $d \geq 2$ keep a legal swap in the *preserve* mode iff $d \notin D(a) \triangle D(b)$; in the *after-birth* mode iff that holds or d occurs in none of $D(x_0), \ldots, D(x_i)$. Both modes delete edges, so

$$\#\text{classes(unrestricted)} \leq \#\text{classes(after birth)} \leq \#\text{classes(preserve)}$$

(proved: deleting edges cannot decrease the component count; the after-birth edge set contains the preserve edge set). Restricting by all indices at once leaves no edge (Lemma 11.2); a single index d is the reduced, informative property.

r	hist.	classes	pres. 2	pres. 3	pres. 4	pres. 8	aft. 2	aft. 3	pres. 12 / 36
1	4	3	4	4	4	3	4	3	3 / 3
2	100	22	36	51	62	54	32	24	43 / 33
3	3 601	142	413	551	864	1 009	355	239	678 / 493

Table 3: **Swap classes of the complete histories of 72^r under persistence restriction (finite-exact; recomputed in this work).** "pres. d" and "aft. d" are the numbers of classes in the preserve and after-birth modes for the index d. The unrestricted counts are $3, 22, 142$.

Proposition 11.5 (finite-exact: persistence restriction is strictly finer and not monotone in d)**.** *(1) Restricted connectivity is strictly finer than unrestricted already at $r = 1$: for $d = 2$ the histories h_1, h_2 of 72 (swap-equivalent) are separated, 4 classes instead of 3. (2) The number of classes is not monotone along divisibility of d: for $r = 3$ the index $d = 12$ gives 678 classes and $d = 36$ gives 493, though $12 \mid 36$ (for $r = 2$: 43 and 33). (3) After-birth is coarser than preserve.* FINITE-EXACT *for $r \le 3$; no counting law in r is known.*

Source record: `OCCURRENCE_BRIDGE:Results` (`CW-0358`); status FINITE-EXACT; treatment FINITE.

11.3 Churn

For a history of length l put $V_d = \sum_i |\rho_d(i+1) - \rho_d(i)|$ and $\chi_d = \frac{1}{2}\big(V_d - |\rho_d(l) - \rho_d(0)|\big)$, the *reduced churn* (the number of lost-and-regained or gained-and-lost excursions).

Theorem 11.6 (churn bounds)**.** *(i) $\chi_d \le \lfloor l/2 \rfloor$. (ii) A swap changes one entry of ρ_d, so $|\chi_d(h) - \chi_d(h')| \le 1$ for swap-adjacent h, h'. (iii) If $\Delta_p(N) \ge 0$ for all labels N of h and primes $p \mid d$, then ρ_d is nondecreasing and $\chi_d = 0$ (for instance $h_{\mathrm{A}} = 1^7$). (iv)* Dip lemma: *if v_p dips to at least 2 below its start value a and ends at or above a, then $\chi_{p^{\lfloor a/2 \rfloor}} \ge 1$.*

Proof. (i) $V_d \le l$ and $V_d \equiv |\rho_d(l) - \rho_d(0)| \pmod 2$, so $2\chi_d \le l$. (ii) Changing one interior entry of a 0/1 sequence changes V_d by at most 2 and leaves the end values alone. (iii) v_p is then nondecreasing for $p \mid d$, so ρ_d is. (iv) With $k = \lfloor a/2 \rfloor$: the start has $v_p = a \ge 2k$, so $\rho = 1$; at the dip $v_p \le a - 2 < 2k$ (as $2k \ge a - 1$), so $\rho = 0$; the end has $v_p \ge a \ge 2k$, so $\rho = 1$; hence $V \ge 2$ and $\chi \ge 1$. □

Source record: `B-T7` (`CW-0389`); status PROVED; treatment FULL.

Theorem 11.7 (necessary churn of the merger)**.** *For every source m and every admissible arrangement of $\{2^2, 3^4\}$ some d has $\chi_d \ge 1$, whereas $\chi_d(1^7) = 0$ for all d. Hence every endpoint-preserving rewrite between 1^7 and an arrangement of $\{2^2, 3^4\}$ creates an occurrence gap.*

Proof. The walk v_3 starts at $b = v_3(m)$ and has steps $+2$ (twice, labels 2) and -1 (four times, labels 3), ending at b, and stays ≥ 0. Take $d = 3^k$, so $\rho_d = [v_3 \ge 2k]$. *Case 1:* the walk reaches $b + 2$. Put $k = \lfloor b/2 \rfloor + 1 \ge 1$; then $2k \ge b + 1 > b$ and $2k \le b + 2$, so $\rho = 0$ at the start and the end, and $\rho = 1$ at the moment of reaching $b + 2$: $\chi \ge 1$. *Case 2:* the walk stays $\le b + 1$. Its first step cannot be $+2$, so it is -1 and the walk reaches $b - 1$ (so $b \ge 1$). If b is even put $k = b/2 \ge 1$: $\rho = 1$ at start and end ($v_3 = b = 2k$) and $\rho = 0$ at $b - 1$. If b is odd and the walk reaches $b - 2$ or lower put $k = (b - 1)/2$: start and end have $\rho = 1$ ($b \ge 2k$), the low point has

$v_3 \le b-2 < 2k$. If b is odd and the walk never goes below $b-1$, it can leave $b-1$ only by a $+2$ step, so it visits $b+1$; with $k=(b+1)/2$, $\rho=0$ at the start and end and $\rho=1$ at $b+1$. In each case $V_d \ge 2$ and $\chi_d \ge 1$. For 1^7 every $\Delta_p(1) \ge 0$, so Theorem 11.6(iii) gives $\chi_d = 0$. □

Source record: `B-T8` (`CW-0390`); status PROVED; treatment FULL.

Check. All arrangements for $v_2(m) \le 8$, $v_3(m) \le 6$ (690 admissible cases) and the monotone and dip statements (1126 and 2162 cases) were replayed with no violation. In the block cube family from 4 the total churn at source 4 equals the number of B-blocks; the minimum number of churning indices over all sources and arrangements within the tested index set is 5 (cited).

11.4 Information across rewrites

Proposition 11.8 (births minus losses is a rewrite invariant)**.** *Let* births *and* losses *be the numbers of index-birth and index-loss events along a history. Then* births − losses $= |D(x_l)| - |D(x_0)|$ *depends only on the endpoint and the source, so it is a rewrite invariant. The fibre sequences are* not *rewrite invariants (rewrites change lengths). In the cube family of Theorem 9.8,* A *has* $(\text{births}, \text{losses}) = (7,0)$ *and* B *has* $(14,7)$*, with difference* 7 *each; the* 2^r *histories of the cube family are pairwise distinguished by their fibre sequences for* $r \le 4$*, while the pairs* $(\text{births}, \text{losses})$ *take* $2, 4, 8, 15, 24$ *distinct values for* $r = 1, \dots, 5$*, so that the pair is a faithful label only for* $r \le 3$*. The first statement is proved by telescoping; the numerical statements are* FINITE-EXACT *and cited.*

Source record: `D4-T9` (`CW-0382`); status PROVED; treatment FULL.

The endpoint fixes the image $V_A c$ and leaves the position in the kernel coset free; block rewrites move within that coset, and a telescoping invariant such as births − losses depends on the endpoint alone and so cannot see that position.

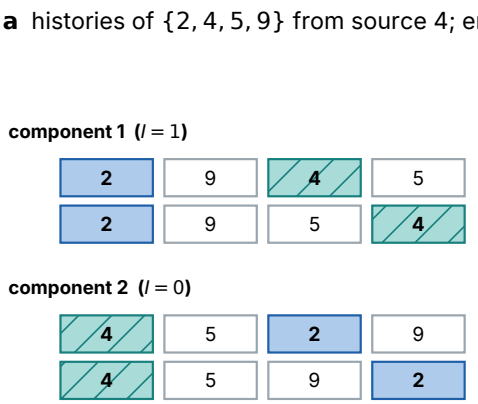
11.5 A separator of swap classes by one occurrence index

Theorem 11.9 (the order of 2 and 4; proved-scoped)**.** *Let* A *be a set of distinct labels with* $A \cap \{1,2,4\} = \emptyset$*, let the source be* 4 *and the content* $C = \{2,4\} \cup A$*. For an admissible arrangement* w *of* C *put* $I(w) = 1$ *if* 2 *precedes* 4 *and* $I(w) = 0$ *otherwise. Then* I *is constant on swap classes. Consequently the number of swap classes is at least the number of values of* I *that occur.*

Proof. A swap exchanging two labels other than the pair $\{2,4\}$ does not change the relative order of 2 and 4. A legal swap of 2 and 4 requires them to be adjacent with $8 = 2 \cdot 4$ dividing the state x before the pair (Proposition 4.5). If the pair is not at the first two positions, the first label of the history lies in A; but at the source 4 the only applicable labels are the divisors $1, 2, 4$ of 4, and A contains none of them. So the pair occupies the first two positions and $x = 4$, and $8 \nmid 4$. □

Derived allocation: `CW-0413/0414 (proposed co-located proof)` (`R3-T3`, outside the 62 allocated rows); status PROVED-SCOPED; treatment FULL.

Example 11.10 (the family from endpoint 9000)**.** For $A = \{5,9\}$ the content $\{2,4,5,9\}$ ends at 9000 and has 2 swap classes, each of 2 histories: $\{(2,9,4,5),(2,9,5,4)\}$ with $I = 1$ and $\{(4,5,2,9),(4,5,9,2)\}$ with $I = 0$ (Figure 19). For $A = \{3,5,9\}$ (endpoint 48000) there are



b finite-exact rows

A	endpoint	comp.	I-values
{5, 9}	9000	2	{0, 1}
{5, 9, 3}	48000	3	{0, 1}
{5, 9, 3, 6}	392000	3	{0, 1}

I = [2 before 4] is constant on every swap class. It is not a complete invariant: for $A = \{3, 5, 9\}$ there are 3 components but I takes two values.

Figure 19: **The order of 2 and 4 separates swap classes (finite-exact data, theorem exact).** (a) The four histories of the content $\{2, 4, 5, 9\}$ from source 4 (endpoint 9000) grouped into the two swap classes; the labels 2 (light fill) and 4 (hatched) are marked, with $I = [2 \text{ before } 4]$. (b) The three rows computed: the sets A, endpoints, number of components and values of I; for $A = \{3, 5, 9\}$ there are 3 components but I takes two values only. Theorem illustrated: Theorem 11.9.

3 classes (sizes $1, 3, 4$) with I-values $1, 1, 0$, and for $A = \{3, 5, 6, 9\}$ (endpoint 392000) there are 3 classes (sizes $5, 8, 14$) with the same I-values: I separates the pair of values but is not a complete invariant. In a sweep over all 39 subsets $A \subseteq \{3, \dots, 12\}$ of size 1 to 3 the invariance of I holds in all cases (0 violations); the number of classes is 1 for 24 contents, 2 for 14 and 3 for 1, and both values of I occur for 14. FINITE-EXACT. The classification of the classes beyond I is not given.

Remark 11.11 (a corrected statement). The square-root content $\sigma(x) = \prod_p p^{\lfloor v_p(x)/2 \rfloor}$ has $D(x)$ as its set of divisors ≥ 2, and along every SELECT-II edge $\sigma'/\sigma = (N+1)/c > 1$ with $c \leq N$ the product of the prime powers $p^{\lfloor \cdot/2 \rfloor}$ or $p^{\lceil \cdot/2 \rceil}$ taken from N (the programme's B2-T9, "no silent step"). The inclusion $D(x) \subsetneq D(x')$ that was once stated with it does *not* hold, as the accepted erratum records: $4 \to 18$ has $D(4) = \{2\}$ and $D(18) = \{3\}$. The occurrence fibre is therefore not monotone along histories, which is why churn is a meaningful quantity. CITED.

12 Relation to individual-state geometry

Manuscript I [1] attached to a single state a geometry; here only the integer part is needed. Two facts about it are taken from the programme's releases and are CITED, not re-proved.

- *Collapse of integer state geometry (B2-T1).* For an integer x put $\sigma(x) = \prod_p p^{\lfloor v_p(x)/2 \rfloor}$. The complete geometric record of x is a function of $\sigma(x)$ alone, equivalent to the occurrence set $D(x)$ (the divisors ≥ 2 of $\sigma(x)$); all its topological invariants are trivial and the move

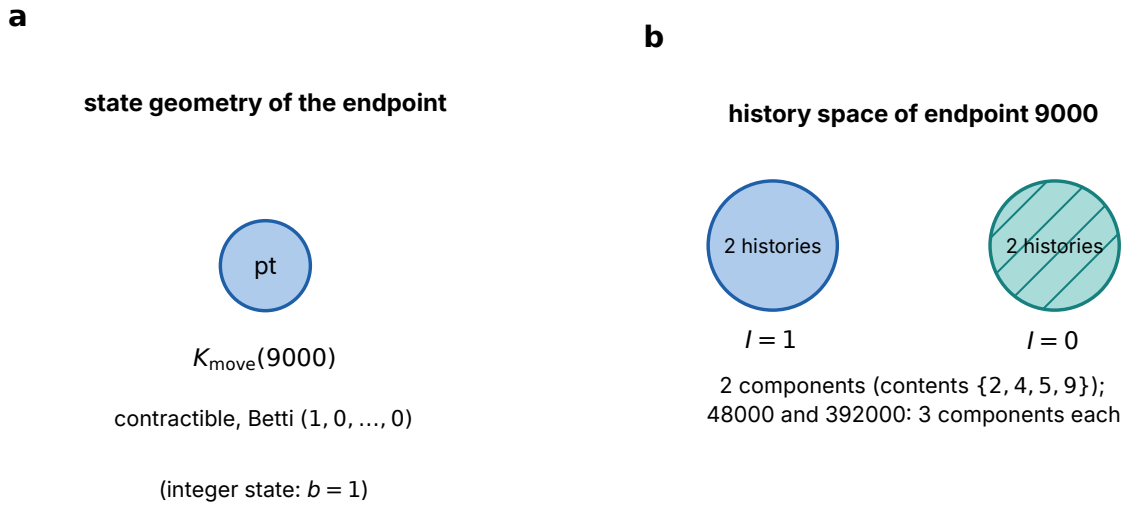


Figure 20: **Contractible state geometry against a disconnected history space (schematic, data finite-exact).** (a) The state geometry of the endpoint 9000 is a point up to homotopy: $K_{\mathrm{move}}(9000)$ is contractible with Betti vector $(1, 0, 0, 0)$ (cited). (b) The swap-class structure of the histories of $\{2, 4, 5, 9\}$ from source 4 ending at 9000: two components of two histories each, distinguished by I (exact). The picture is schematic; no map between (a) and (b) is claimed or constructed. Theorem illustrated: Proposition 12.1.

complex $K_{\mathrm{move}}(x)$ of an integer state is contractible. What varies between integers is dimension, shape, the f-vector and the inclusion order.

- *Different objects.* $K_{\mathrm{move}}(x)$ lives on the occurrence set of one state; the history complexes of this manuscript live on admissible words from a fixed source. They are not identified.

Proposition 12.1 (obstruction to detecting histories by state homology)**.** *Every integer endpoint E of a SELECT-II history has $K_{\mathrm{move}}(E)$ contractible (cited). Hence no homological invariant of $K_{\mathrm{move}}(E)$ can detect the component structure of the histories ending at E. Witnesses: the endpoints* 9000*,* 48000*,* 392000 *have* 2*,* 3*,* 3 *components of histories with contents $\{2, 4\} \cup A$, $A = \{5, 9\}, \{5, 9, 3\}, \{5, 9, 3, 6\}$ (source* 4*; recomputed in this work, Example 11.10), and the order bit $I = [2$ before $4]$ is constant on components and takes both values (Theorem 11.9); each $K_{\mathrm{move}}(E)$ has $\mathbb{F}_2$-Betti vector $(1, 0, 0, 0)$ (cited).* CITED *for the contractibility;* FINITE-EXACT *for the component counts.*

Derived allocation: `GEO_HD_BRIDGE_v004:A4-T7 (obstruction)` (`A4-T7`, outside the 62 allocated rows); status CITED; treatment CITED.

The figure above is Figure 20; its caption states the exactness of what is drawn. What is *not* established. No functor, map or inequality between $H_*(K_{\mathrm{move}})$ and the homology of a history complex is claimed. A direct comparison would need an invariant of rational states that is meaningful for histories without extending SELECT-II, which the model boundary of this

manuscript excludes; the question whether such an invariant exists is open (Section 13). The only integer-level connection proved here is the occurrence-event one: the fibre sequence of a history is built from $D(x_i) = D(\sigma(x_i))$ (Theorem 11.3), so the state geometry of the visited states determines the fibre sequence while the state geometry of the endpoint alone does not (Proposition 12.1).

13 Synthesis, conclusions and open problems

13.1 Mathematical conclusions

1. *The endpoint forgets the history.* For the endpoint 72 the four histories have three contents and three swap classes (Figure 1). In general, for a fixed source and alphabet, the endpoint determines the image $V_A c$, i.e. the content only modulo $\ker_{\mathbb{Z}} V_A$ (Theorem 9.1), and says nothing about order.

2. *Order is governed by one divisibility.* Admissibility of a word is $K(w) \mid m$ (Theorem 4.2); every swap, class and component statement of Sections 5 and 6 is a statement about the induced subgraph of the arrangement graph on $\{w : K(w) \mid m\}$.

3. *Same content, different classes.* Resource contention, not numerical incompatibility, separates arrangements: $(14, 3, 2)$ and $(2, 3, 14)$ commute numerically and are not connected by admissible swaps at 14 (Theorem 5.8); this is the hexagon deadlock of Theorem 6.7.

4. *Topology.* The permutohedral route complex has the homology of the nerve of the infeasible arrangements (Theorem 7.3); complementary infeasible pairs give the homology of S^{k-3} (Theorem 7.5), realised by a loop for $\{2, 3, 8, 9\}$ at 72 and a void for $\{2, 3, 8, 9, 14\}$ at 1008. Cube dimension is not homological degree (Remark 8.5), and the dimension of an obstruction in terms of prime resources is not the rank of the valuation span (Proposition 8.8).

5. *Rewrites.* Different-content histories with the same endpoint are the integer kernel of V_A, all kernel vectors are realised, a rewrite is local to a prefix state (Theorem 10.2), and below the saturation threshold three obstruction types occur; the exposure threshold of the family $2^4 3^j \leftrightarrow 1 \cdot 8^2 3^j$ is exactly $3 \mid m$ or $16 \mid m$ for every j (Theorem 10.8).

6. *Occurrence events.* The fibre sequence determines the history (Theorem 11.3); the merger $1^7 \leftrightarrow 2^2 3^4$ necessarily creates occurrence gaps (Theorem 11.7); the pair order of 2 and 4 separates swap classes at endpoints $9000, 48000, 392000$ although the state geometry of those endpoints is contractible (Proposition 12.1).

13.2 Corrections and sharpenings made in this manuscript

- The all-arrangements statement is an exact equivalence, $P(C) \mid m$ (Proposition 4.6); the converse for *connectivity* is false ($\{2, 3, 14\}$ at 28 is connected though $2 \cdot 3 \cdot 14 = 84 \nmid 28$).
- The sphere statement is homological: complementary infeasible pairs give the homology of S^{k-3}, the homotopy type is not proved (Theorem 7.5).
- The area potential changes by $-2\, d_a \wedge d_b$ under an adjacent exchange, not $d_a \wedge d_b$ (Theorem 9.5); the $\{2^2, 3^4\}$ formula $-7(8 - 2\,\mathrm{inv})$ is consistent only with the factor 2.

- The exposure dichotomy is proved for every j (Theorem 10.8); the source's finite range $j \leq 8$ is superseded.
- B2-T9 originally stated an inclusion $D(m) \subsetneq D(m')$, which fails ($4 \to 18$); the correct statement is the strict increase of σ (Remark 11.11).
- *Status of the* $\{2, 3, N\}$ *converse.* In the source release the arithmetic part of D-T9 is proved and the converse (a split source exists only if $N \equiv 2, 3 \pmod 6$) is finite-exact for $4 \leq N \leq 400$. In this manuscript the converse is proved for every $N \geq 4$: Theorem 6.9(a) shows by the block factorisation Theorem 6.3 that $\nu \leq 1$ unless the feeding digraph is strongly connected, i.e. unless $N \equiv 2, 3 \pmod 6$, and Theorem 6.9(b) classifies the split sources in the strongly connected case by Theorem 6.7. The source's finite range is retained as corroboration only.
- Cited, not re-run: the census of $k \leq 5$ labels (cut order 1 in about 17 900 instances), the loop converse for $4 \leq N < M \leq 60$, the 498 of 816 deficient fibres, the births-and-losses table.

13.3 Open problems

Problem 13.1 (swap-class classification)**.** Classify the swap classes of all contents, beyond the contents of one to three labels and the family $\{2, 3, N\}$: the structure of the lcm-closure $L(C)$ and the maximal number of components for k labels. OPEN (D-Q1).

Source record: `D-Q1` (`CW-0513`); status OPEN; treatment OPEN.

Problem 13.2 (common futures)**.** Unbounded statements on common futures of histories (two histories with a common continuation) are known only as bounded regression data. OPEN (D-Q3).

Source record: `D-Q3` (`CW-0515`); status OPEN; treatment OPEN.

Problem 13.3 (image of the fibre-sequence map)**.** Which sequences of fibres occur as fibre sequences of histories? Theorem 11.3 shows injectivity but not the image. OPEN (D-Q4).

Source record: `D-Q4` (`CW-0516`); status OPEN; treatment OPEN.

Problem 13.4 (overlapping cells)**.** The Independence Lemma fills cubes for non-overlapping exchanges only (Remark 8.4). A complex with cells for overlapping moves, the effect of coherence cells on the extended complex of Problem 10.11, and 319 triples with pairwise diamonds but no triple cell for $m < 400$ (cited) are open. OPEN (D-Q5).

Source record: `D-Q5` (`CW-0517`); status OPEN; treatment OPEN.

Problem 13.5 (interface to the next stage)**.** An invariant of rational states meaningful for histories without extending SELECT-II (Section 12); the relation of the history complexes to the programme's later stages. OPEN (D-Q6).

Source record: `D-Q6` (`CW-0518`); status OPEN; treatment OPEN.

Further problems stated in the text: growth of $H(72^r)$ and of the maximal cube dimension (Problem 3.6); whether K_m is always a wedge of spheres (Problem 7.13); complexes for

repeated labels (Problem 8.11); homotopy type and Künneth behaviour of extended complexes (Problem 10.11); synergistic obstructions of cut order ≥ 2 (Proposition 8.10); a proof of connectivity for $\{1, 2, 3\}$ at all sources (Proposition 10.6); parallel displacement vectors for all N (Proposition 9.7).

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[2] UNNS Substrate programme, *Mathematical Foundations of the UNNS Substrate, Volume I: Generative Arithmetic and Structural Reachability* (the grammar SELECT-II and the occurrence set). Source: `ukbbi/UNNS`, commit `9a021346b2b3d964523440d2737b9d0c4ddbb3e5`, `projects/UNNS_MATH_FOUNDATIONS/Manuscript`.

[3] UNNS Substrate programme, *Mathematical Foundations of the UNNS Substrate, Volume II: Collective Dynamics, Prime Rigidity, and Effective Predictability*. Same commit and path as [2]. Cited only for the existence of the rational dynamics FIELD-I, which this manuscript does not use.

[4] Foundations reference collection v001 (SELECT-I/II, FIELD-I v001–v012; frozen), `ukbbi/UNNS`, commit `bb8b12d753c600c597feb00df17fe139c7cc65fa` (23 Oct. 2026), `projects/UNNS_MATH_FOUNDATIONS/sources`.

[5] GEO-HD-1 research collection, snapshot `a88f31487f3b05c9db297dcd1192d56eea2b0c7d` (the pin named in the programme specification; not independently verified in this work). The release archives from which the records of this manuscript are taken, with their SHA-256 as computed on the files supplied for this work:

- `GEO_HD_1D_v001` `c532b7410f3bf9d437f9f172b3e56029cd9f7bab5b2ac0c0e533713155eaf7a4`;
- `GEO_HD_1D_v002` `8baf463c07bb5285dd53b355aee05a0f841c068ba3ae4794f3702adff6a4bae0`;
- `GEO_HD_1D_v003` `65dbb281e4c75d625b6cf3e4bcb8262d7d42e68d0c476fbba225fcde744a4706`;
- `GEO_HD_1D_v004` `e0390e2a2f88eaeb11042de09fd056741ed98f94318de708d308faa090f05b5b`;
- `GEO_HD_BRIDGE_v001` `e99472809928f4a23fb292f60759965159c7a4db5f018d56748ccac328719395`;
- `GEO_HD_BRIDGE_v002` `012ead97e5b8854cd5218417d46f0e388d927b6d59d5f6e75a8cc343f5dd4955` (with the accepted erratum `GEO_HD_BRIDGE_v002_ERRATUM_B2-T9.md`);
- `GEO_HD_BRIDGE_v003` `6fba87af9f3096772e43f4ac77fa8d55aea9e242eeef241081a987b567d5d5a8`;
- `GEO_HD_BRIDGE_v004` `cfb31157a39666d1a4bbb33b280ec626c7de2c141e8232da43c8b934b8afffff4`;
- `GEO_HD_BRIDGE_v005` `50e342821d7a5f760c5127df348e97f3a4d4d3bf34bf60e1019d6336f28c5d03`.

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On the references. UNNS sources are cited at the commits and archive hashes fixed by the programme; repository access during this work was read-only and the pinned repositories were not fetched from the environment in which the manuscript was prepared. The external literature is cited by author, title and publication data. It is cited only for *resemblance* of structure (commutation classes, rewriting systems, Markov bases, directed topology) and for the standard theorems named; nothing in the manuscript is proved by appeal to an equivalence with those theories.

A Source theorem crosswalk

A.1 Provenance records

Every result of this manuscript that restates a record of the programme's crosswalk carries a line "*Source record*" in the text. The table below is generated at build time from these lines (`scripts/make_appmap.py`); it lists the crosswalk row, the source identifier, the status recorded in the crosswalk, the proof treatment of the allocation (FULL: proof given here; FINITE: exact finite computation, presented as such; OPEN: stated as a problem; SPLIT: proved part and finite-exact part stated separately), and the section and number at which it appears. Rows appearing twice are restated in two places.

CW row	source id	status	treatment	section	number
A4-T7	GEO_HD_BRIDGE_v004:A4-T7 (obstruction)	cited	cited	12	12.1
CW-0336	D-T0	PROVED	FULL	3	3.1
CW-0337	D-W	PROVED	FULL	2	2.4
CW-0338	D-T1	PROVED	FULL	4	4.5
CW-0339	D-T2	PROVED	FULL	3	3.4
CW-0340	D-T3	PROVED	FULL	8	8.1
CW-0341	D-T4	PROVED	FULL	11	11.3
CW-0342	D-T5	PROVED	FULL	5	5.9
CW-0343	D-T6	FINITE-EXACT	FINITE	8	8.4
CW-0344	D-T7	PROVED	FULL	4	4.2

CW row	source id	status	treatment	section	number
CW-0345	D-T7’	PROVED	FULL	5	5.2
CW-0346	D-T8	PROVED	FULL	5	5.5
CW-0347	D-T9	MIXED	SPLIT	6	6.10
CW-0348	D-T10	PROVED	FULL	5	5.8
CW-0349	THEOREMS_V002:Census-FINITE-EXACT-results	FINITE-EXACT	FINITE	6	6.12
CW-0350	NF	PROVED	FULL	6	6.2
CW-0351	F2	PROVED	FULL	6	6.3
CW-0351	F2	PROVED	FULL	6	6.4
CW-0352	Hub-descent	PROVED	FULL	6	6.6
CW-0353	CONNECTIVITY:Census-on-three-labels-FINITE	FINITE-EXACT	FINITE	6	6.6
CW-0354	T3	PROVED	FULL	6	6.7
CW-0355	OBSTRUCTIONS:Parameter-family-2-3-N-PROV	PROVED	FULL	6	6.10
CW-0356	OCCURRENCE_BRIDGE:Orthant-form	PROVED	FULL	11	11.1
CW-0357	OCCURRENCE_BRIDGE:Local-discrepancy	PROVED	FULL	11	11.4
CW-0358	OCCURRENCE_BRIDGE:Results	FINITE-EXACT	FINITE	11	11.5
CW-0359	T-DUAL	PROVED	FULL	7	7.3
CW-0360	T-LOWER	PROVED	FULL	7	7.4
CW-0361	T-SPHERE	PROVED	FULL	7	7.5
CW-0362	ROUTE_TOPOLOGY:Cubical-versus-permutohedral	FINITE-EXACT	FINITE	7	7.10
CW-0363	ROUTE_TOPOLOGY:Census-FINITE-EXACT-contents	FINITE-EXACT	FINITE	7	7.12
CW-0364	ROUTE_TOPOLOGY:Statement-of-the-general-conje	OPEN	OPEN	7	7.13
CW-0365	D6-T1	PROVED	FULL	4	4.2
CW-0366	(P) Prefix-set form	PROVED	FULL	4	4.3
CW-0367	(G) Dependence on the source	PROVED	FULL	4	4.4
CW-0368	D6-T2	PROVED	FULL	4	4.7
CW-0369	D4-T1	PROVED	FULL	9	9.1
CW-0370	D4-T2	PROVED	FULL	9	9.2
CW-0371	D4-T3	PROVED	FULL	10	10.2
CW-0372	D4-T4	PROVED	FULL	9	9.3
CW-0373	D4-T5	PROVED	FULL	9	9.8
CW-0374	D4-T6a	PROVED	FULL	10	10.3
CW-0375	D4-T6b	PROVED	FULL	10	10.4
CW-0376	D4-T6c	FINITE-EXACT	FINITE	10	10.6
CW-0377	D4-T7a	PROVED	FULL	10	10.9
CW-0378	D4-T7b	MIXED	SPLIT	10	10.9
CW-0379	D4-T7c	PROVED	FULL	10	10.7
CW-0380	D4-T8a	PROVED	FULL	10	10.10
CW-0381	D4-T8b	OPEN	OPEN	10	10.11
CW-0382	D4-T9	PROVED	FULL	11	11.8
CW-0383	B-T1	PROVED	FULL	8	8.7
CW-0384	B-T2	PROVED	FULL	8	8.8
CW-0385	B-T3	PROVED	FULL	8	8.9
CW-0386	B-T4	FINITE-EXACT	FINITE	8	8.10
CW-0387	B-T5	PROVED	FULL	9	9.5
CW-0388	B-T6	FINITE-EXACT	FINITE	9	9.7
CW-0389	B-T7	PROVED	FULL	11	11.6
CW-0390	B-T8	PROVED	FULL	11	11.7
CW-0391	B-T9	PROVED	FULL	10	10.8
CW-0513	D-Q1	OPEN	OPEN	13	13.1
CW-0514	D-Q2	OPEN	OPEN	3	3.6
CW-0515	D-Q3	OPEN	OPEN	13	13.2
CW-0516	D-Q4	OPEN	OPEN	13	13.3
CW-0517	D-Q5	OPEN	OPEN	13	13.4
CW-0518	D-Q6	OPEN	OPEN	13	13.5
R3-T3	CW-0413/0414 (proposed co-located proof)	proved-scoped	full	11	11.9

A.2 Allocation notes

- *Accepted transfer.* The records B-T1 to B-T9 (CW-0383 to CW-0391) are history and rewrite theory carried by the bridge releases; their transfer from the stage-III allocation to this

manuscript is recorded in `crosswalk/PAPER_II_DERIVED_ALLOCATION.md`.

- *Proposed co-location.* R3-T3 (`CW-0413`, `CW-0414`), the separator $I = [2 \text{ before } 4]$, is restated with a complete proof in Theorem 11.9; its allocation remains with the stage-III manuscript, and the proof here is a proposed co-located one (printed as "derived allocation").
- *Cited, not re-proved.* B2-T9 (as corrected by the accepted erratum) in Remark 11.11; B2-T1 and A4-T7 in Section 12. No identification of state geometry with history topology is claimed.
- *Mixed rows.* D-T9 ($\{2, 3, N\}$: the arithmetic statement proved, the converse finite-exact in the source; here the converse is proved for all $N \geq 4$ by the factorisation theorem and the three-label classification, Theorem 6.9, and the source's range $N \leq 400$ is retained as corroboration) and D4-T7b (superseded for all j by B-T9, Theorem 10.8).
- *Release pins.* Archives and commits are listed in the references [5, 2, 4].

B Complete arithmetic examples

All tables of this appendix are generated by `scripts/make_appB_data.py` with the independent exact engine `scripts/engine.py` and are reproduced verbatim from its output (exact integer arithmetic, no floating point). Section numbers in the headings below refer to this manuscript.

B.1 The four complete histories of 72

source	word	states	content
1	$(1, 1, 2)$	$1 \to 4 \to 16 \to 72$	$1, 1, 2$
1	$(1, 2, 1)$	$1 \to 4 \to 18 \to 72$	$1, 1, 2$
3	$(3, 2)$	$3 \to 16 \to 72$	$2, 3$
10	(5)	$10 \to 72$	5

B.2 Thresholds of the arrangements of $\{2, 3, 14\}$

arrangement w	$K(w)$
$(2, 3, 14)$	14
$(2, 14, 3)$	28
$(3, 2, 14)$	21
$(3, 14, 2)$	21
$(14, 2, 3)$	28
$(14, 3, 2)$	14

B.3 Complexes of Section 7 and Section 8

$\{2, 3, 8, 9\}$ **at** 72 **(loop)** Admissible arrangements: 16 of 24. Swap classes: 1. Betti vector of K_m over $\mathbb{F}_2$: $(1, 1, 0)$; f-vector $(16, 20, 4)$.

$\{2, 3, 8, 9, 14\}$ **at** 1008 **(void)** Admissible arrangements: 96 of 120. Swap classes: 1. Betti vector of K_m over $\mathbb{F}_2$: $(1, 0, 1, 0)$; f-vector $(96, 180, 98, 12)$.

$\{2,3,4,5,6\}$ **at** 4 **(Proposition 8.9)** Admissible arrangements: 10 of 120. Swap classes: 2. Betti vector of K_m over $\mathbb{F}_2$: $(2,0,0)$; f-vector $(10,9,1)$. Component sizes: 4, 6.

$\{2,3,4,5,6,7\}$ **at** 4 **(Proposition 8.9)** Admissible arrangements: 24 of 720. Swap classes: 2. Betti vector of K_m over $\mathbb{F}_2$: $(2,0,0)$; f-vector $(24,29,7)$. Component sizes: 12, 12.

$\{2,3,4,5,8\}$ **at** 32 **(Proposition 8.9)** Admissible arrangements: 30 of 120. Swap classes: 1. Betti vector of K_m over $\mathbb{F}_2$: $(1,1,0)$; f-vector $(30,41,11)$. Component sizes: 30.

$\{2,3,4,5,6,7\}$ **at** 24 **(Proposition 8.9)** Admissible arrangements: 87 of 720. Swap classes: 1. Betti vector of K_m over $\mathbb{F}_2$: $(1,1,0,0)$; f-vector $(87,151,70,6)$. Component sizes: 87.

B.4 The overlap hexagon

The content $\{1,2,4\}$ from source 2: admissible arrangements $(1,2,4)$, $(1,4,2)$, $(2,1,4)$ of 6; $P(C) = 8 \nmid 2$.

B.5 The dimensions of an obstruction (Proposition 8.8)

For $C = \{2,3,8,9\}$ at 72 the displacement vectors over the primes $2,3,5$ are $\Delta(2) = (-1,2,0)$, $\Delta(3) = (4,-1,0)$, $\Delta(8) = (-3,4,0)$ and $\Delta(9) = (2,-2,2)$ (since $10 = 2 \cdot 5$); the rank is 3. For $C = \{2,3,8,9,14\}$ at $1008 = 2^4 \cdot 3^2 \cdot 7$ the label 14 adds $\Delta(14) = (-1,2,2,-1)$ in the primes $(2,3,5,7)$ ($15 = 3 \cdot 5$), and the rank over $2,3,5,7$ is 4. The prime 7 is consumed once, by the unique label 14, and $v_7(1008) = 1$, so it never constrains; relaxing 2 or 3 changes the admissible set and relaxing 7 does not. Keeping only the constraints at $\{2,3\}$ reproduces the Betti vector $(1,0,1,0)$ of the full complex, and neither $\{2\}$ nor $\{3\}$ alone does.

B.6 The kernel merger and the cube family

Source 4: 1^7 has states $4, 16, \ldots, 65536$; $(2,2,3,3,3,3)$ has states $4, 18, 81, 432, 2304, 12288, 65536$; $(2,3,2,3,3,3)$ and $(2,3,3,2,3,3)$ reach the same endpoint (Example 9.4). The block cube family has 2^r histories ending at 4^{7r+1}, e.g. $r = 1,2,3$: 65536, 1073741824, 17592186044416 $(= 4^8, 4^{15}, 4^{22})$, replayed in this work.

C History-complex construction

C.1 The permutohedral route complex

Let $C = \{N_1, \ldots, N_k\}$ be distinct labels and $\mathcal{A}(m) \subseteq S_k$ the set of admissible arrangements. The boundary complex ∂P_k of the permutohedron [15] has as faces the ordered set partitions $(B_1|\ldots|B_s)$ of $\{1,\ldots,k\}$, $s \geq 2$; its vertices are the $k!$ arrangements (all blocks singletons) and a face contains an arrangement when the arrangement lists the blocks in order. The *route complex* K_m is the full subcomplex on $\mathcal{A}(m)$: a face is present iff every arrangement it contains is admissible. The cubical complex Q_m instead has as cells the sets of pairwise non-overlapping legal swaps of one history (Lemma 8.1); the two coincide on the 1-skeleton, differ in higher cells, and the hexagon of three labels is a 2-cell of K_m but not of Q_m.

C.2 Chain complexes and the check $\partial^2 = 0$

Both complexes are used with $\mathbb{F}_2$ coefficients and cellular boundary $\partial\sigma = \sum_{\tau\subset\sigma,\ \dim\tau=\dim\sigma-1} \tau$. For the face poset of ∂P_k (a regular CW sphere) and for the cubical complex every codimension-two incidence is a diamond, so $\partial^2 = 0$; in addition the program asserts $\partial_{i-1}\partial_i = 0$ numerically before each rank computation (`scripts/topo.py`, function `betti_from_boundaries`), and the Betti numbers are checked against the Euler characteristic of the f-vector in every table of this manuscript. Those checks passed in every reported case (for instance $16 - 20 + 4 = 0$ and $1 - 1 + 0 = 0$ for the loop of $\{2, 3, 8, 9\}$; $96 - 180 + 98 - 12 = 2$ and $1 - 0 + 1 - 0 = 2$ for the void).

C.3 Implementation

Cells of K_m are enumerated as ordered set partitions whose linear extensions all lie in $\mathcal{A}(m)$; ranks over $\mathbb{F}_2$ are computed by Gaussian elimination on bit-packed rows. For repeated labels no such construction is given (Problem 8.11). General background on complexes of this type: [16, 17].

D Kernel and rewrite calculations

D.1 The valuation matrix of $\{1, 2, 3, 8\}$

Rows $p = 2, 3$, columns $N = 1, 2, 3, 8$:

$$V = \begin{pmatrix} 2 & -1 & 4 & -3 \\ 0 & 2 & -1 & 4 \end{pmatrix},$$

with kernel of rank 2 over $\mathbb{Z}$. Each of the five Graver candidates of Section 9 satisfies $Vz = 0$:

z	2-row	3-row
$(1, -4, 0, 2)$	$2 + 4 + 0 - 6$	$0 - 8 + 0 + 8$
$(2, 5, -2, -3)$	$4 - 5 - 8 + 9$	$10 + 2 - 12$
$(3, 1, -2, -1)$	$6 - 1 - 8 + 3$	$2 + 2 - 4$
$(4, -3, -2, 1)$	$8 + 3 - 8 - 3$	$-6 + 2 + 4$
$(7, -2, -4, 0)$	$14 + 2 - 16$	$-4 + 4$

(each row sums to 0).

D.2 The $\{3^2 8,\ 1^3 2\}$ pair

$R(3)^2 R(8) = \left(\frac{16}{3}\right)^2 \frac{81}{8} = 288 = R(1)^3 R(2) = 64 \cdot \frac{9}{2}$. The difference of the content vectors is $(3, 1, 0, 0) - (0, 0, 2, 1) = (3, 1, -2, -1) = z_3$.

D.3 The family $2^4 3^j \leftrightarrow 1 \cdot 8^2 3^j$

$R(2)^4 = \frac{6561}{16} = \frac{3^8}{2^4}$ and $R(1)R(8)^2 = 4 \cdot \frac{6561}{64} = \frac{3^8}{2^4}$. The valuation walks: a leading 3 adds 4 to v_2; 2 changes (v_2, v_3) by $(-1, +2)$; 8 by $(-3, +4)$; 1 by $(+2, 0)$; 3 by $(+4, -1)$. For $b \geq 1$ the words $3, 2, 2, 2, 2, 3^{j-1}$ and $3, 1, 8, 8, 3^{j-1}$ have v_3 trajectories $b - 1, b + 1, b + 3, b + 5, b + 7, \ldots$ and $b - 1, b - 1, b + 3, b + 7, \ldots$, both ending at $b + 8 - j$.

D.4 The join example

$R(2)R(8)R(63) = \frac{9}{2}\cdot\frac{81}{8}\cdot\frac{4096}{63} = \frac{20736}{7} = \frac{36}{5}\cdot\frac{100}{9}\cdot\frac{1296}{35} = R(5)R(9)R(35)$. At $m = 280$ the histories $(2, 63, 8)$, $(8, 63, 2)$, $(5, 9, 35)$, $(35, 9, 5)$ end at $280 \cdot 20736/7 = 829440$.

E Supplementary proofs

E.1 The threshold is the least admissible source

Claim. The admissible sources of a word w are exactly the multiples of $K(w)$, and $K(w)$ is the least. *Proof.* By Theorem 4.2, w is admissible from m iff $K(w) \mid m$. □

E.2 Euler characteristic of Q for 72^r

The f-vectors $(4, 1)$, $(100, 96, 11)$, $(3601, 6366, 2832, 294, 4)$ give $\chi = 4-1 = 3$, $100-96+11 = 15$ and $3601 - 6366 + 2832 - 294 + 4 = -223$; the Betti vectors (3), $(22, 7)$, $(142, 368, 3)$ give 3, $22 - 7 = 15$, $142 - 368 + 3 = -223$. The agreement is a consistency check on the boundary matrices, not a proof of the Betti numbers (which are exact finite computations).

E.3 Two small proofs used in the text

The pair $\{2, 4\}$ cannot be exchanged except at the start. By Proposition 4.5 the exchange of a, b at a state x is legal iff $ab \mid x$. For $\{2, 4\}$ this requires $8 \mid x$. At the source 4 with every other label outside $\{1, 2, 4\}$ the pair must open the history (Theorem 11.9), $x = 4$ and $8 \nmid 4$.

The invariance of births − losses. Along a history $\text{births}_i - \text{losses}_i = |D(x_i)| - |D(x_{i-1})|$ at step i, because $D(x_i) \setminus D(x_{i-1})$ are the births and $D(x_{i-1}) \setminus D(x_i)$ the losses; summing telescopes to $|D(x_l)| - |D(x_0)|$, which depends on the source and the endpoint only.